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EE6222 Machine Vision

Video (Part I)

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About me

Education:

- Bachelor: Electronic Engineering and Information Science, University of Science and Technology of China
- PhD: EEE, Nanyang Technological University

Research:

- Medical Image Computing
- Robotics
- Computer Vision
- Machine Learning

Project:

- AI-enhanced Synthetic Aperture Radar Imaging with Denoising and Super-resolution for Coastal Projection, 2026-2029
- CARDIA-GM: Cardiac Advanced Recognition for Detection of mIcrovascular Abnormalities – Generalizable Model.
- Towards Realistic Deep Learning for 3D Vision", MTC Programmatic Fund, 2023-2026.
- Learning-based Approach to High Pitch Helical CT Reconstruction and Validation with Cardiac / abdominal CT scan, RCA with General Electric, 2023-2025

https://samjcheng.github.io/

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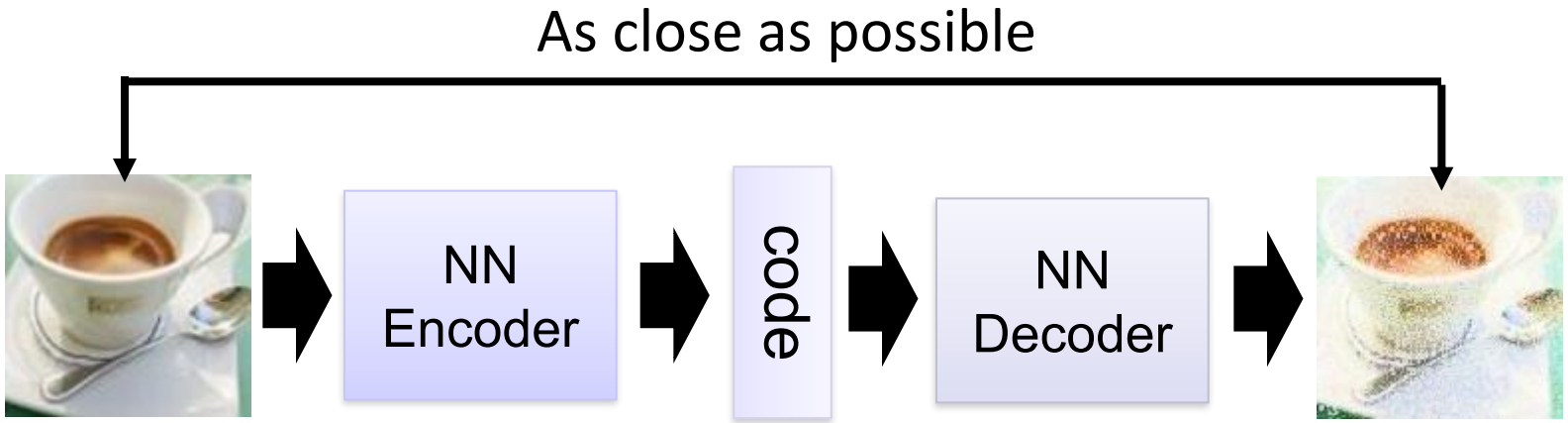
Content

- 1 Video: Generation
- 2 Video: Analysis/Enhancement
- 3 3D: Keypoint and Structure from Motion
- 4 3D: Stereo Matching and Others.

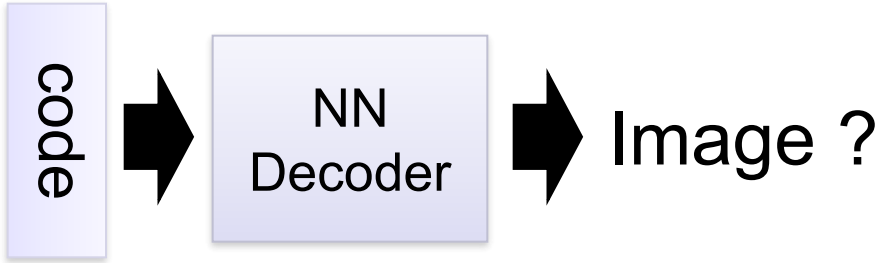
Video Generation


CLICK
BBC
NEWS
WH
.GOV

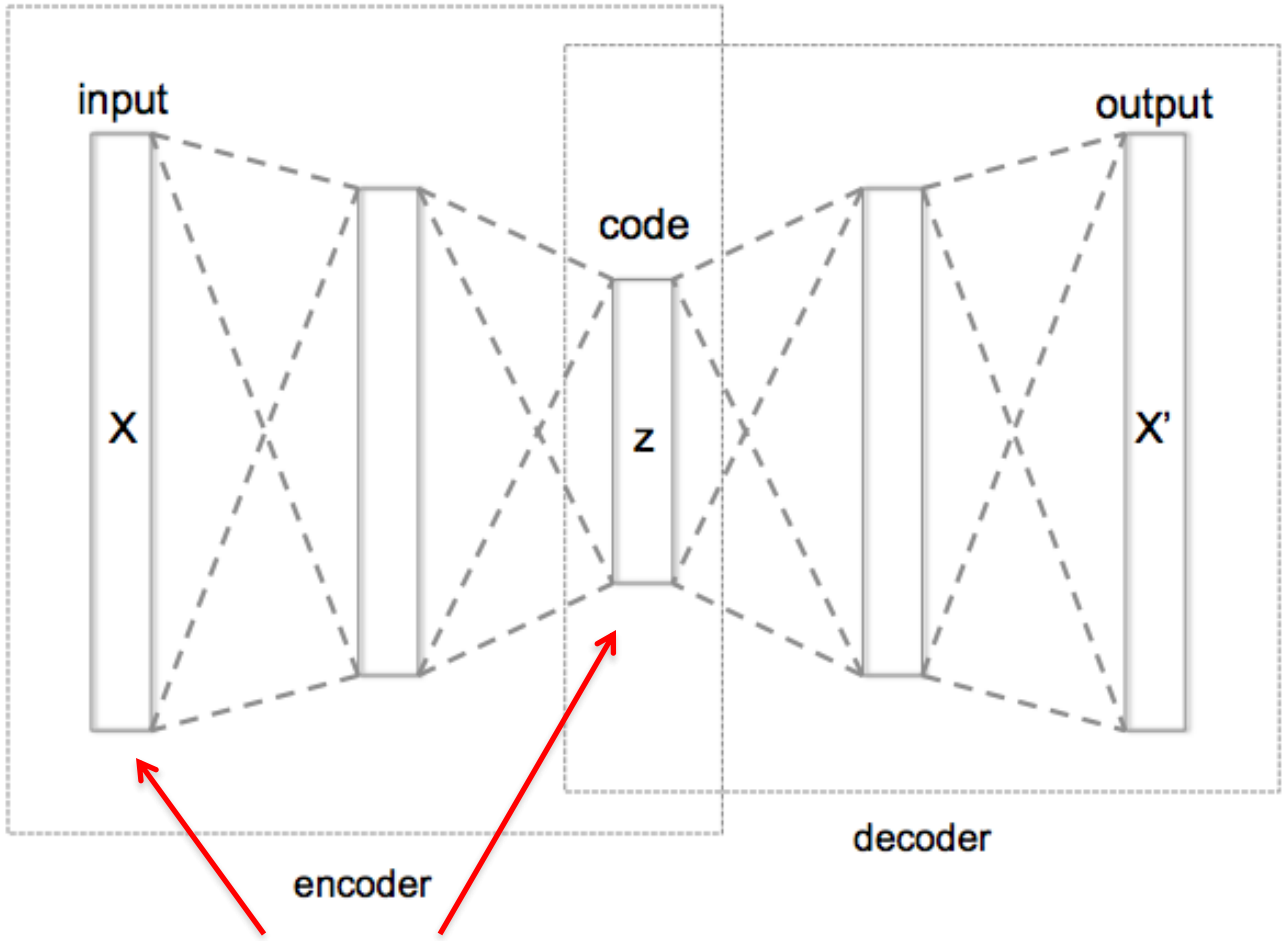
Autoencoder



Randomly generate a vector as code



Autoencoder with 3 fully connected layers

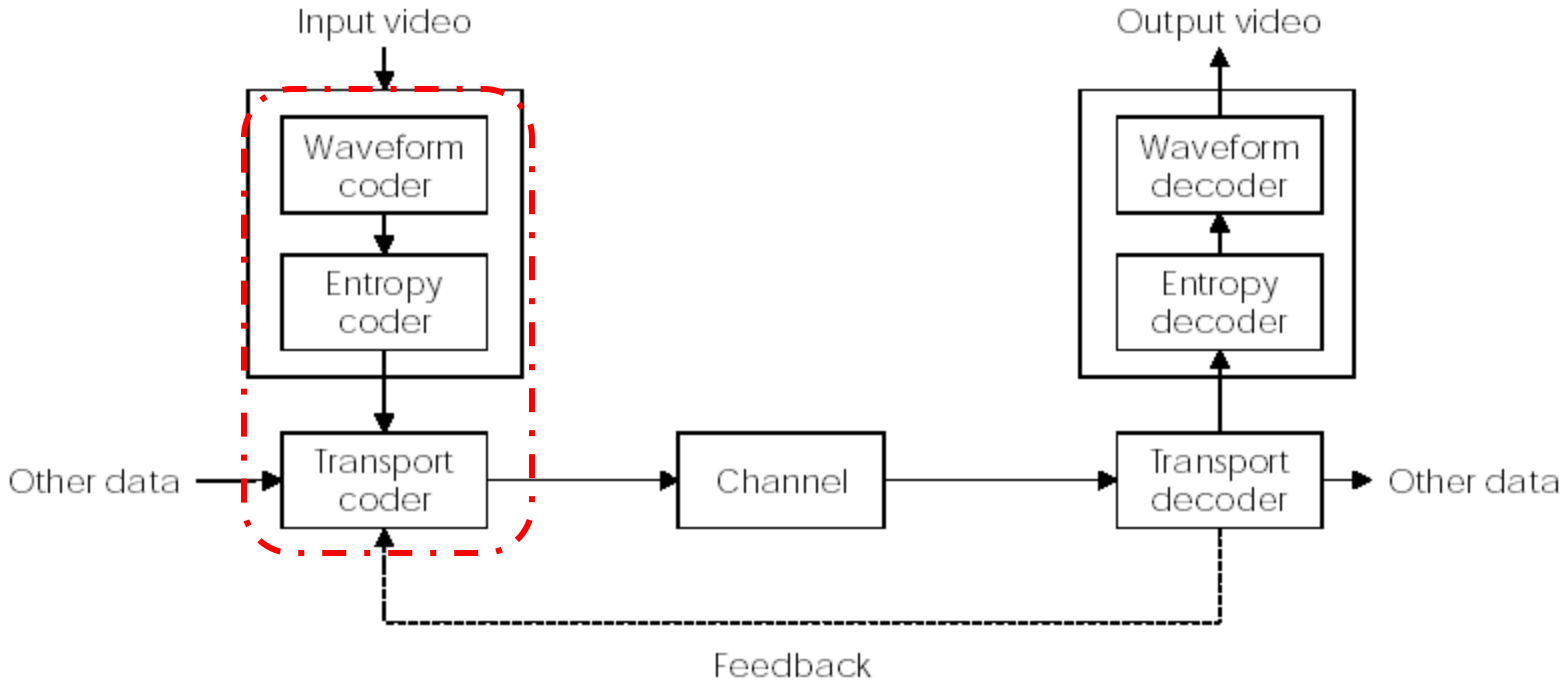


Large → small, learn to compress

Recall that

Video Communication System

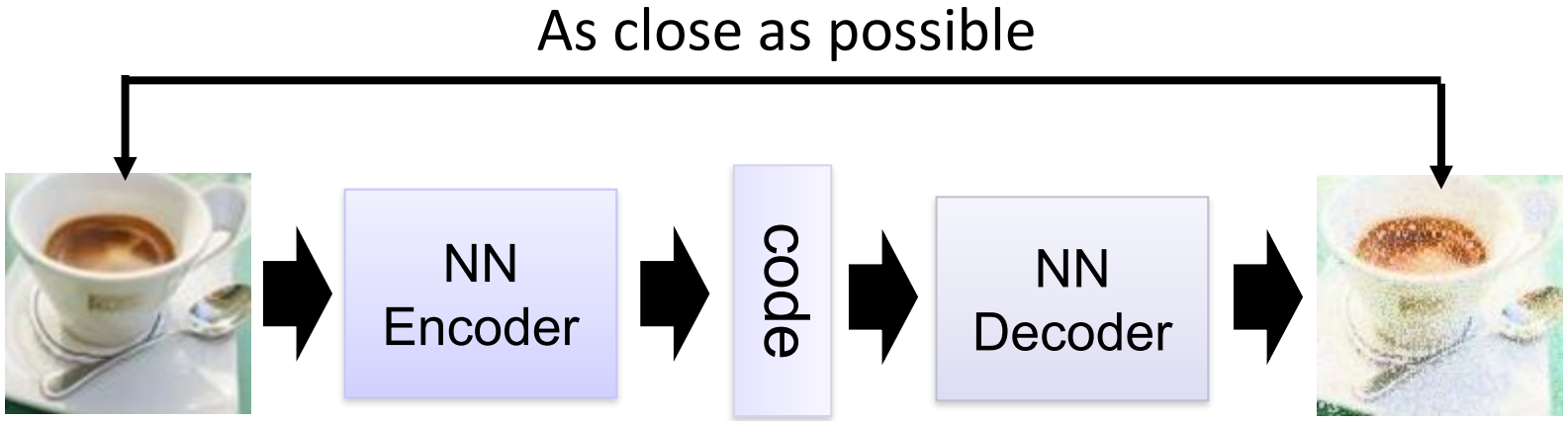
- A functional block diagram for a video communication



A quick test

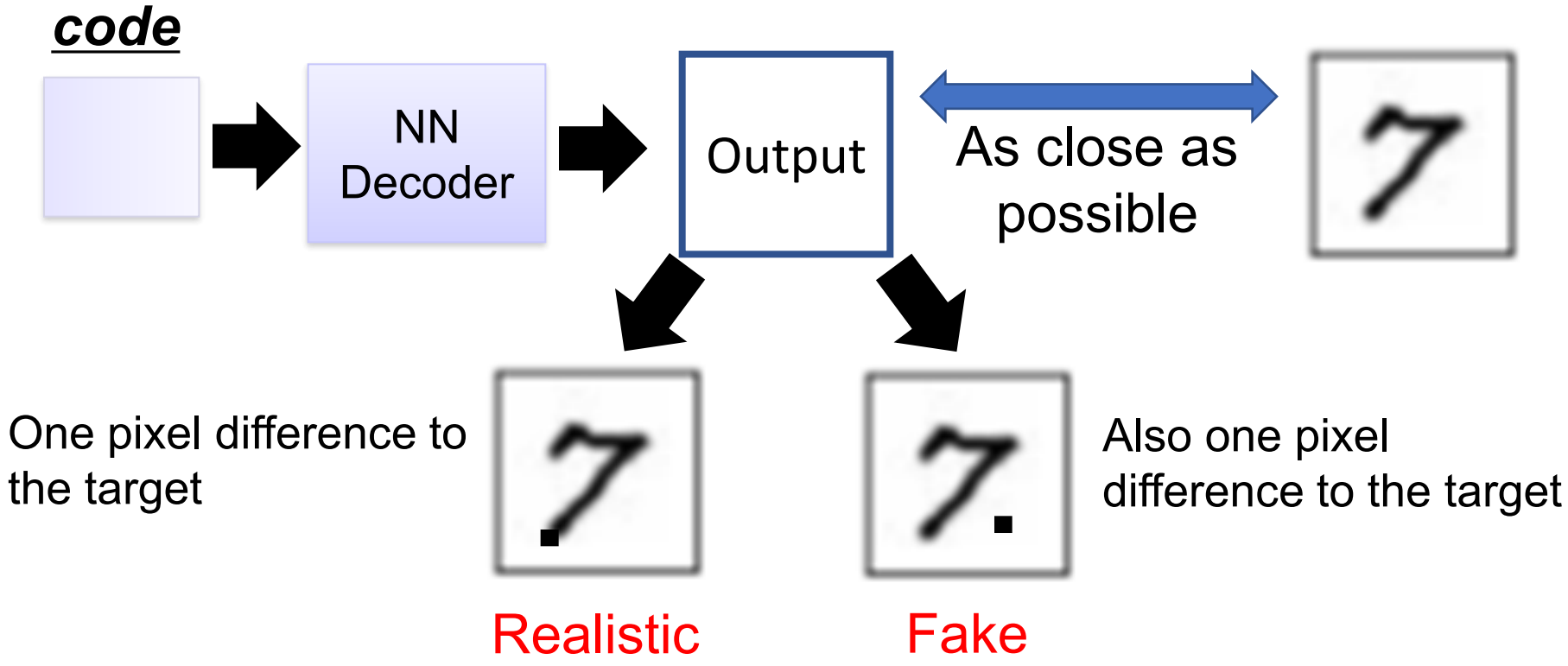
Why we can restore most of the information from the ‘code’

A. The original image has a lot of redundance
B. Trained auto-encoder contain domain knowledge on how to reconstruct the data.

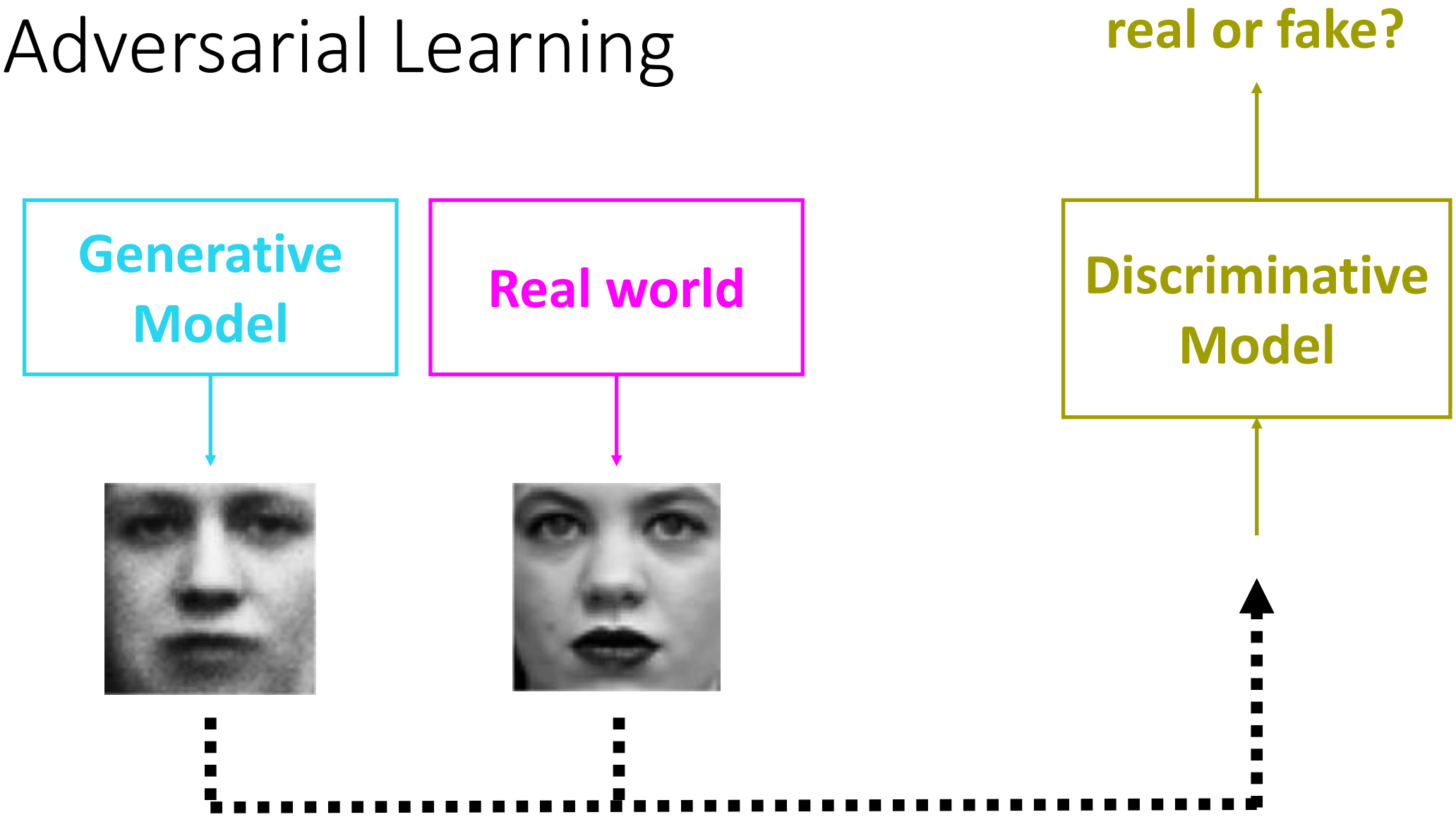


Problems of Autoencoder

- It does not really try to simulate real images



Adversarial Learning

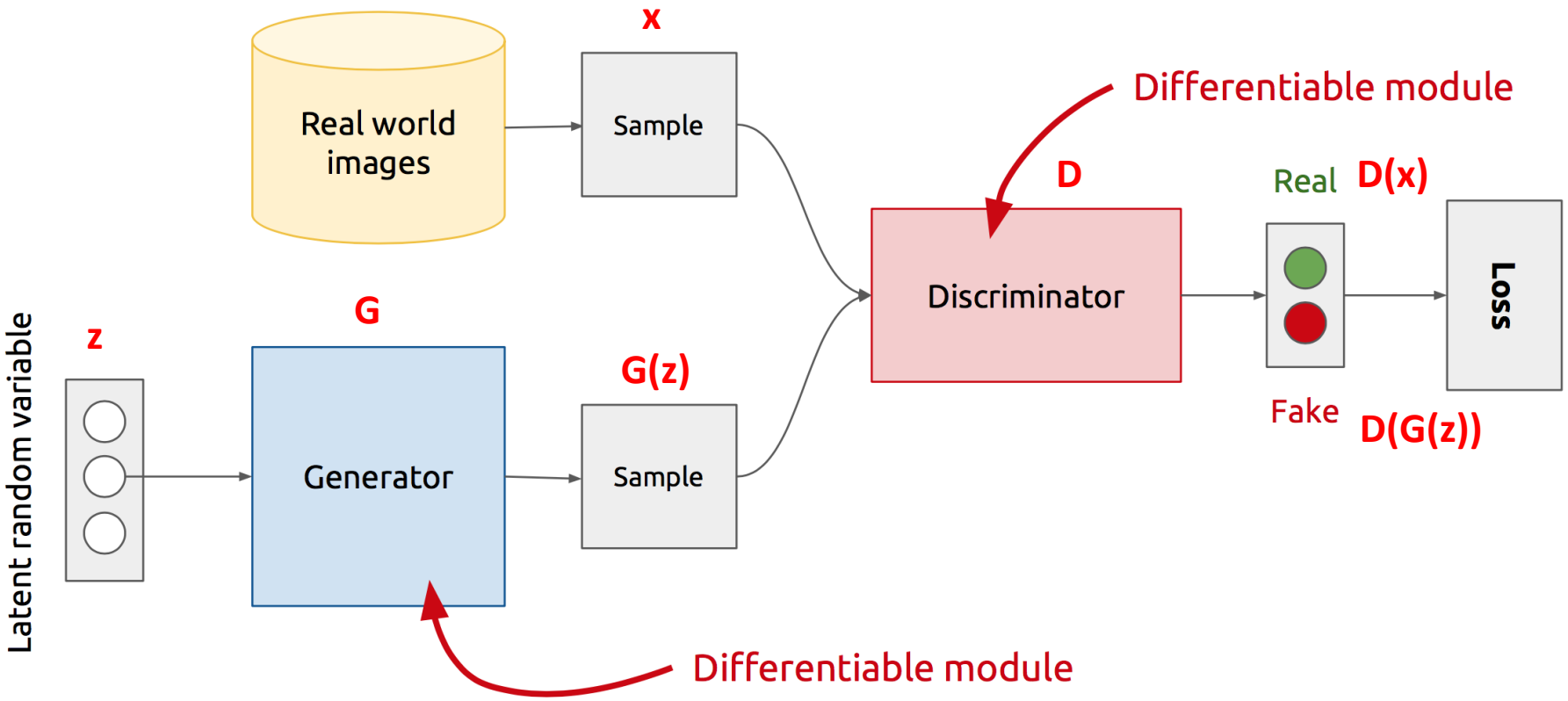


Generative Adversarial Networks (GANs)

- **Generative**
  - Learn a generative model
- **Adversarial**
  - Trained in an adversarial setting
- **Networks**
  - Use Deep Neural Networks

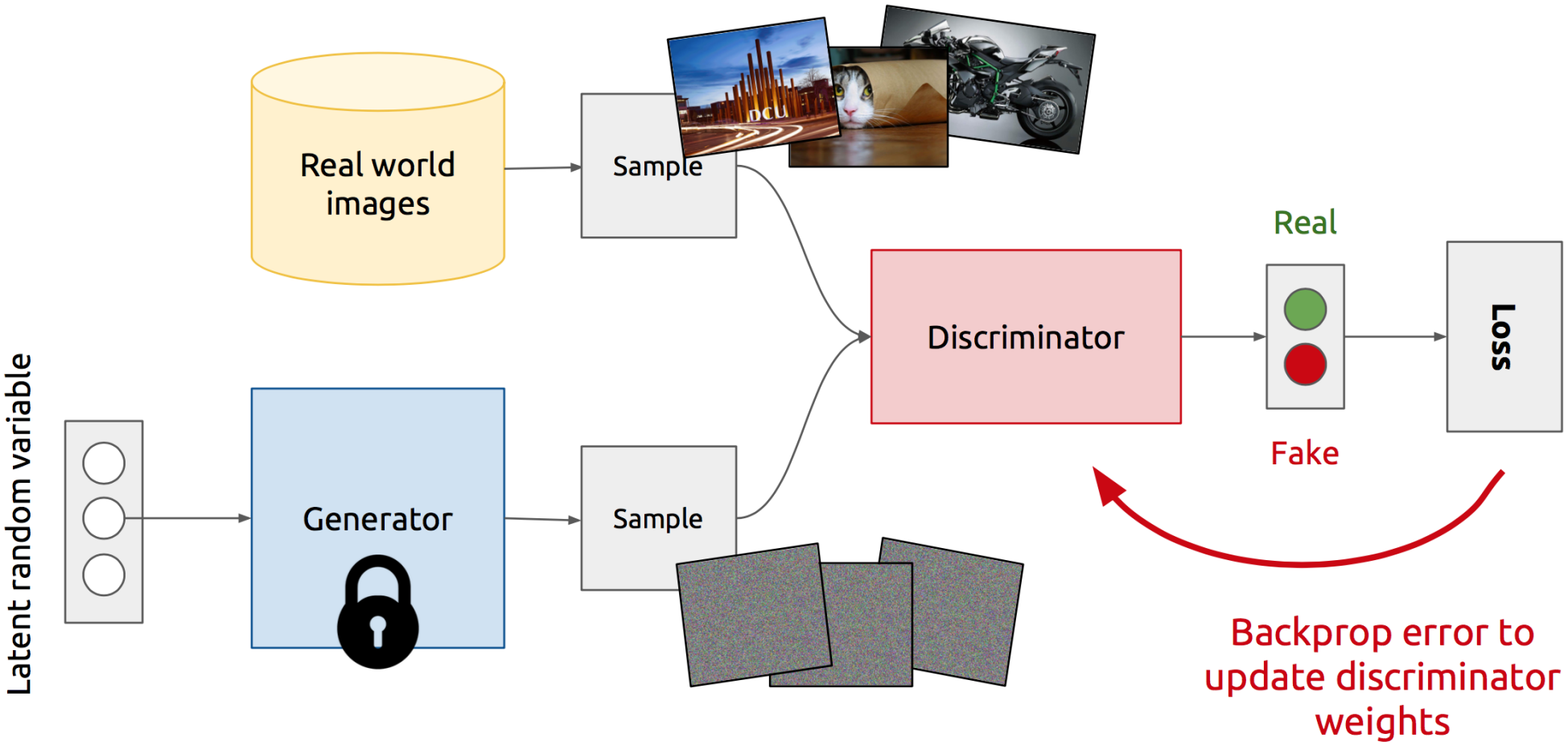
"The most interesting idea in the last ten years in machine learning" LeCun Yann

Structure of GANs

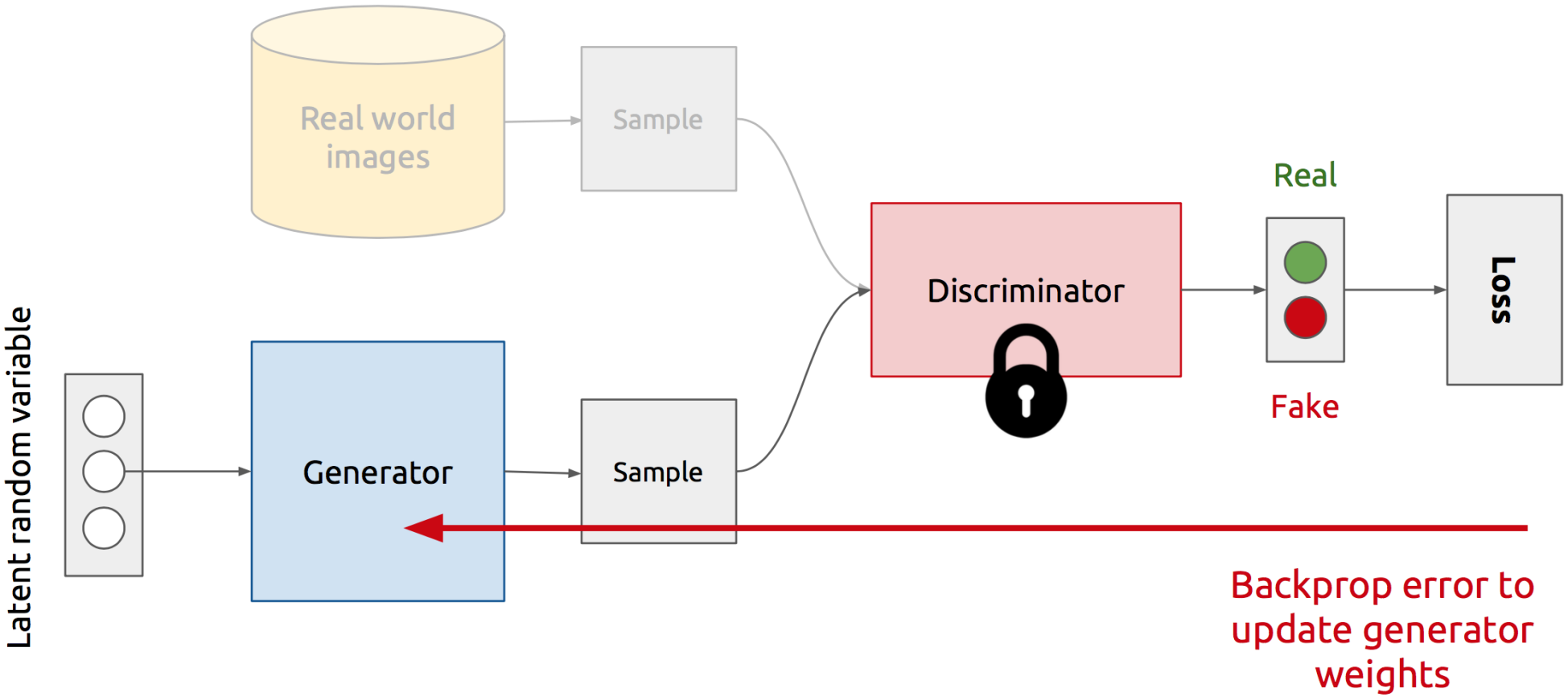


- **Z** is some random noise (Gaussian/Uniform).
- **Z** can be thought as the latent representation of the image.

Training Discriminator



Training Generator



GAN's formulation

$$\min_G \max_D V(D,G)$$

- It is formulated as a **minimax game**, where:
  - The Discriminator is trying to maximize its reward $\boldsymbol{V(D,G)}$
  - The Generator is trying to minimize Discriminator's reward (or maximize its loss)

$$V(D,G) = \mathbb{E}_{x\sim p(x)}[\log D(x)] + \mathbb{E}_{z\sim q(z)}\left[\log\left(1 - D(G(z))\right)\right]$$

- The Nash equilibrium of this particular game is achieved at:
  - $P_{data}(x) = P_{gen}(x) \;\; \forall x$
  - $\mathrm{D}(x) = \frac{1}{2} \;\; \forall x$

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k, is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
- Sample minibatch of m examples $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\boldsymbol{x})$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(\boldsymbol{x}^{(i)}\right) + \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right) \right].$$

end for

- Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

Discriminator updates

Generator updates

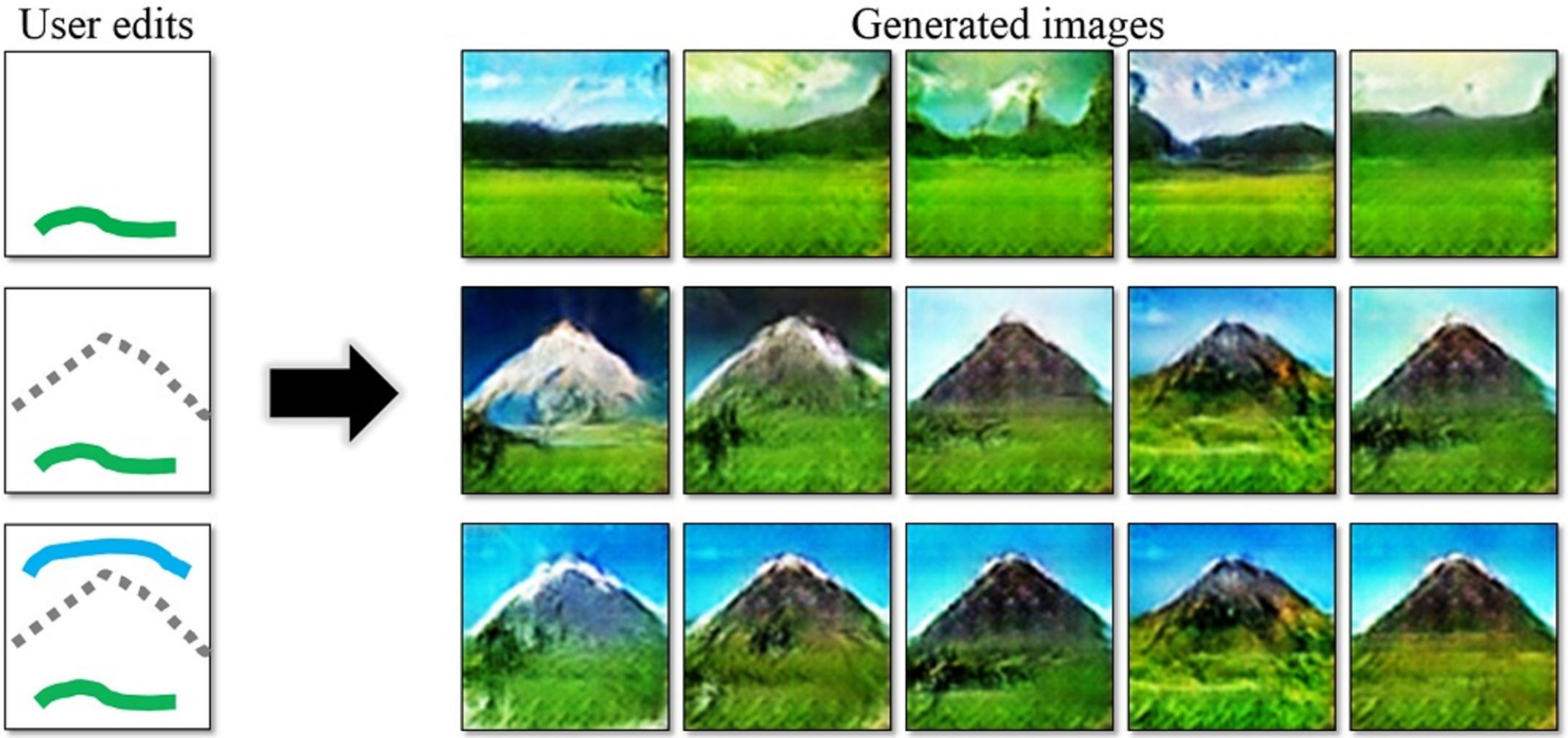
Magic of GANs

Which one is computer generated?





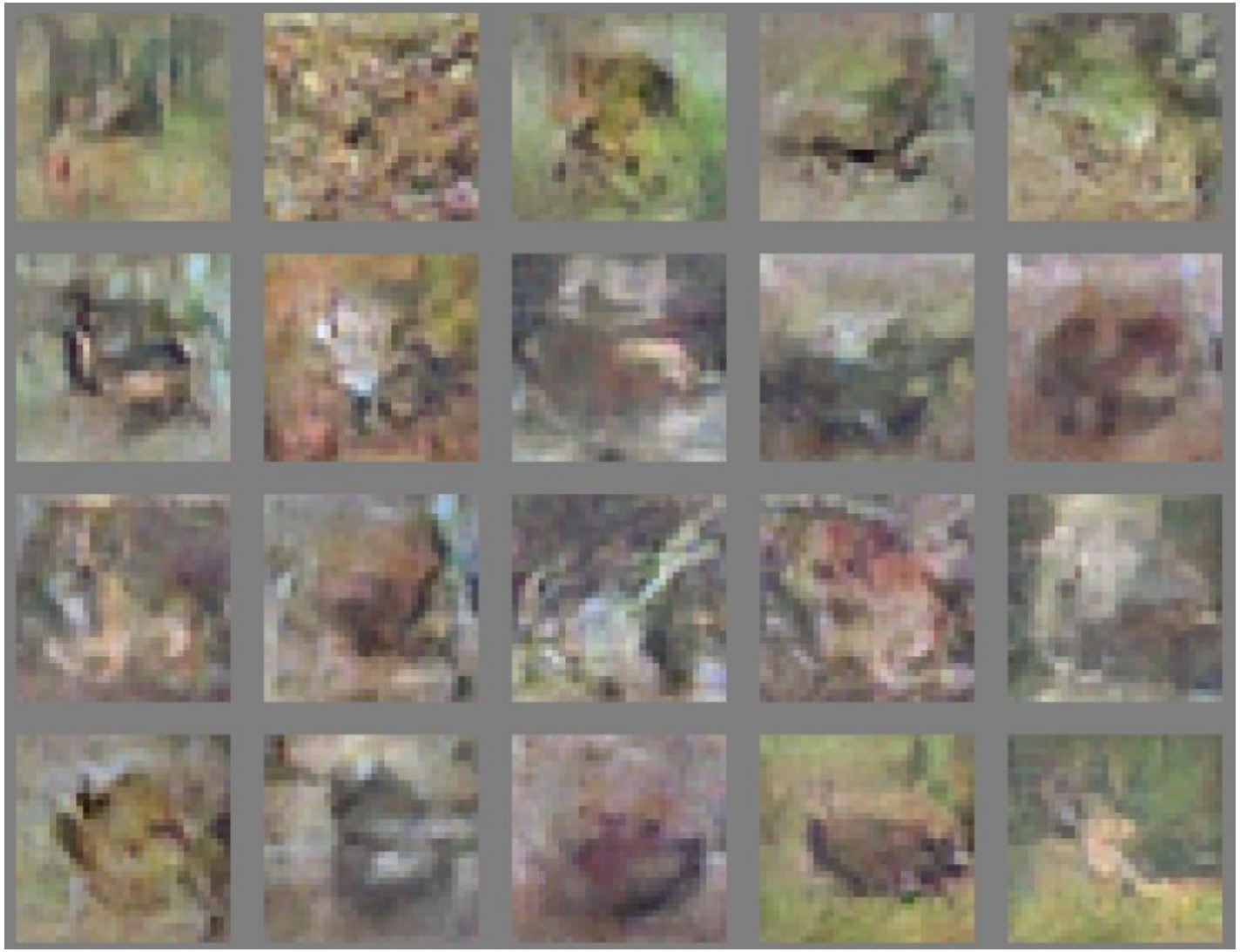
Magic of GANs



Faces



CIFAR

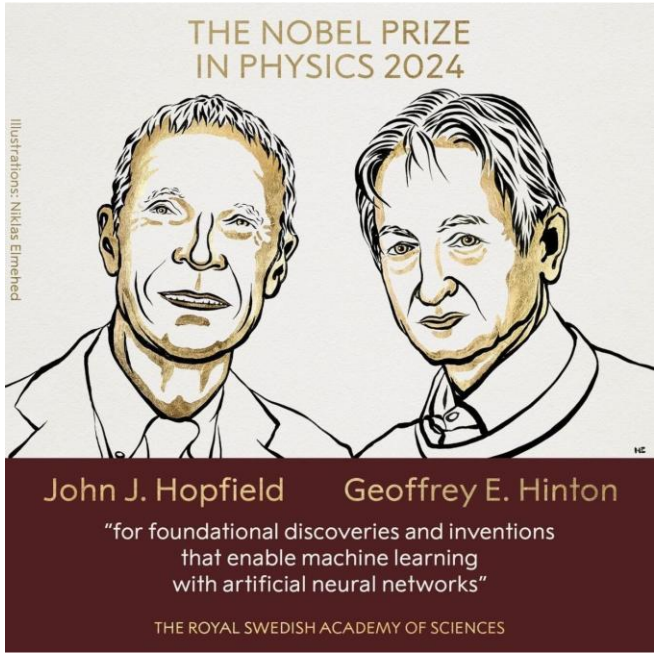


DCGAN



Advantages of GANs

- **Plenty of existing work on Deep Generative Models**
  - Boltzmann Machine
  - Deep Belief Nets
  - Variational AutoEncoders (VAE)



- **Why GANs?**
  - Sampling (or generation) is straightforward.
  - Robust to Overfitting since Generator never sees the training data.
  - Empirically, GANs are good at capturing the modes of the distribution.

Normal model vs GAN

- **Deep Learning models (in general) involve a single player**
  - The player tries to maximize its reward (minimize its loss).
  - Use SGD (with Backpropagation) to find the optimal parameters.
  - SGD has convergence guarantees (under certain conditions).
  - **Problem:** With non-convexity, we might converge to local optima.

$$\min_G L(G)$$

- **GANs instead involve two (or more) players**
  - Discriminator is trying to maximize its reward.
  - Generator is trying to minimize Discriminator's reward.

$$\min_G \max_D V(D, G)$$

  - SGD was not designed to find the Nash equilibrium of a game.
  - **Problem:** We might not converge to the Nash equilibrium at all.

Non-convergence

$$\min_x \max_y V(x, y)$$

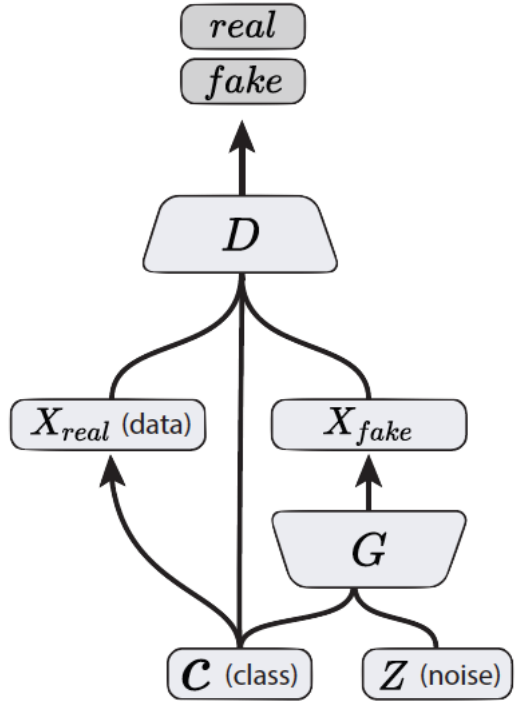
Let $V(x, y) = xy$

• State 1:	x > 0	y > 0	V > 0	Increase y	Decrease x
• State 2:	x < 0	y > 0	V < 0	Decrease y	Decrease x
• State 3:	x < 0	y < 0	V > 0	Decrease y	Increase x
• State 4 :	x > 0	y < 0	V < 0	Increase y	Increase x
• State 5:	x > 0	y > 0	V > 0 == State 1	Increase y	Decrease x

Conditional GANs

- Simple modification to the original GAN framework that conditions the model on *additional information* for better multi-modal learning.

- Lends to many practical applications of GANs when we have explicit *supervision* available.



Conditional GAN
(Mirza & Osindero, 2014)

Image Credit: Figure 2 in Odena, A., Olah, C. and Shlens, J., 2016. Conditional image synthesis with auxiliary classifier GANs. *arXiv preprint arXiv:1610.09585*.

Conditional GANs

MNIST digits generated conditioned on their class label.

$[1, 0, 0, 0, 0, 0, 0, 0, 0, 0] \rightarrow$

$[0, 1, 0, 0, 0, 0, 0, 0, 0, 0] \rightarrow$

$[0, 0, 1, 0, 0, 0, 0, 0, 0, 0] \rightarrow$

$[0, 0, 0, 1, 0, 0, 0, 0, 0, 0] \rightarrow$

$[0, 0, 0, 0, 1, 0, 0, 0, 0, 0] \rightarrow$

$[0, 0, 0, 0, 0, 1, 0, 0, 0, 0] \rightarrow$

$[0, 0, 0, 0, 0, 0, 1, 0, 0, 0] \rightarrow$

$[0, 0, 0, 0, 0, 0, 0, 1, 0, 0] \rightarrow$

$[0, 0, 0, 0, 0, 0, 0, 0, 1, 0] \rightarrow$

$[0, 0, 0, 0, 0, 0, 0, 0, 0, 1] \rightarrow$

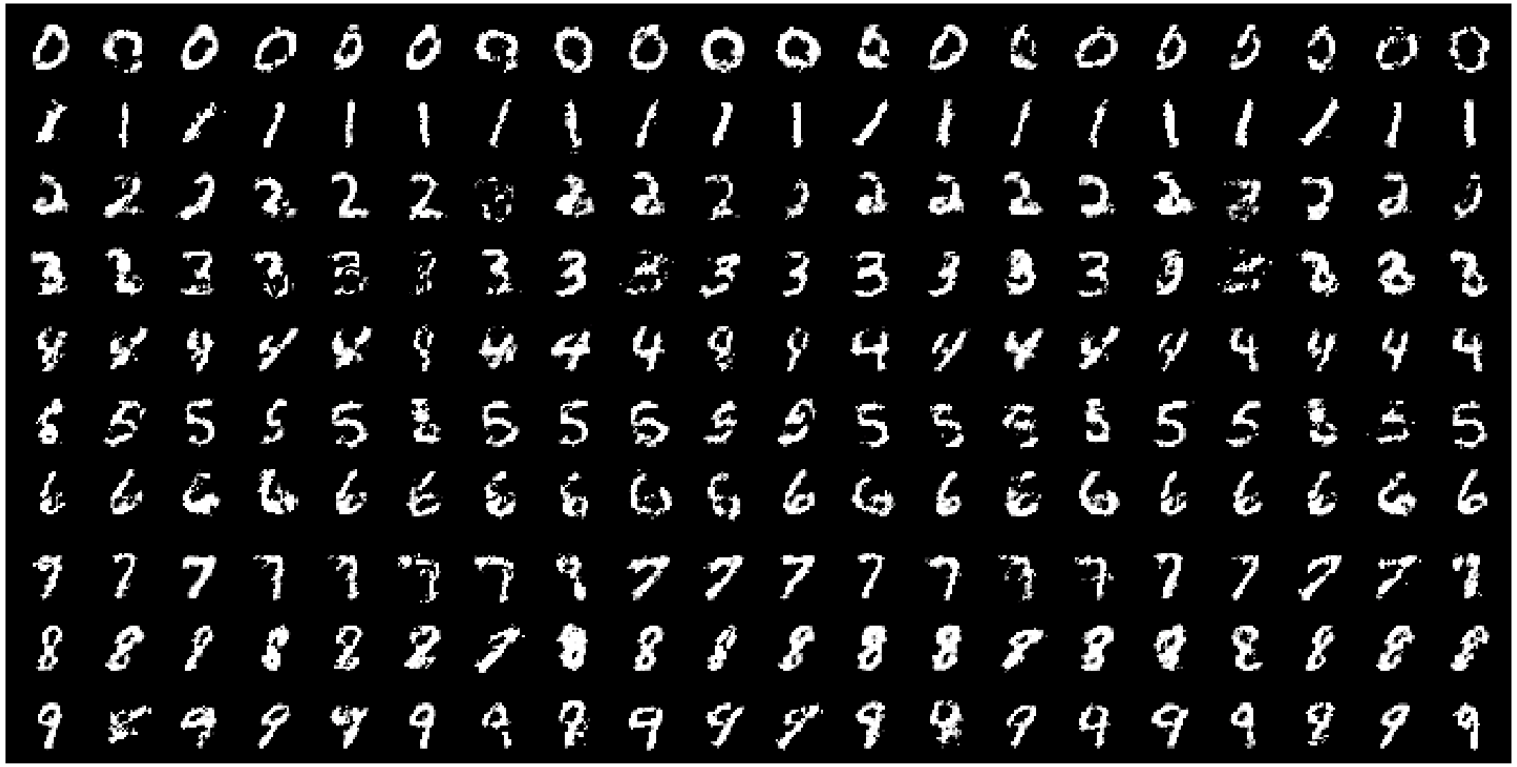
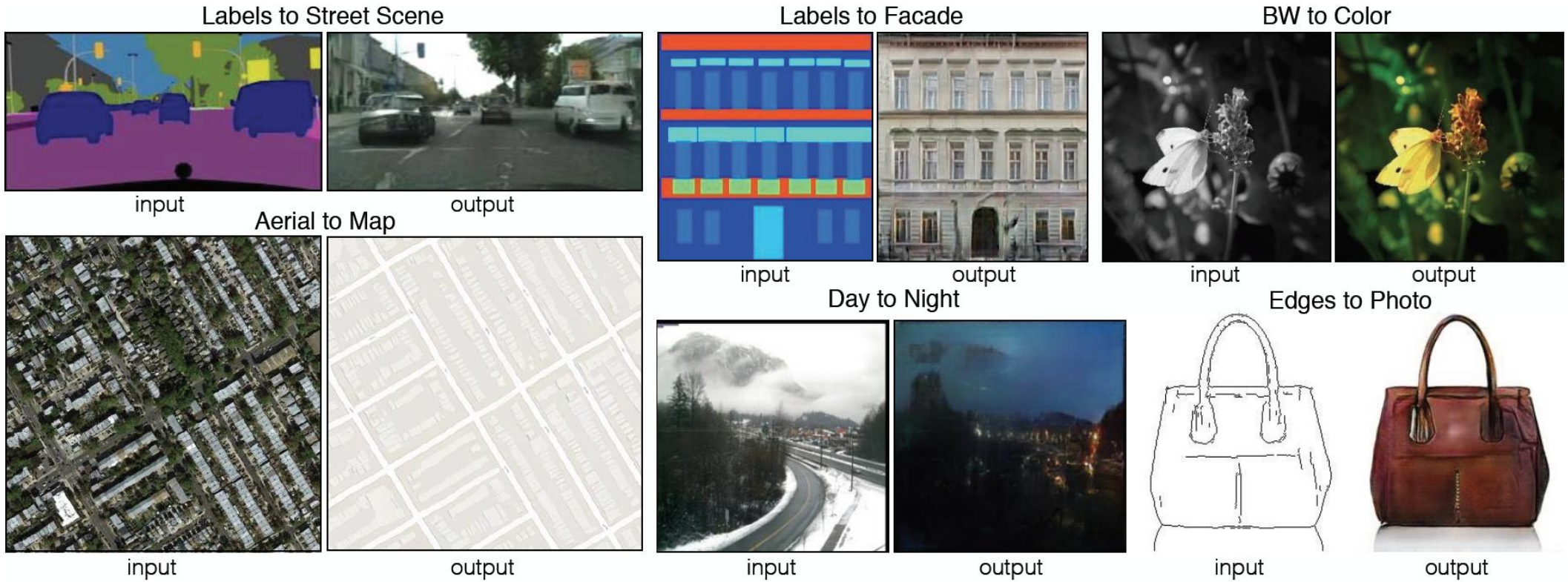
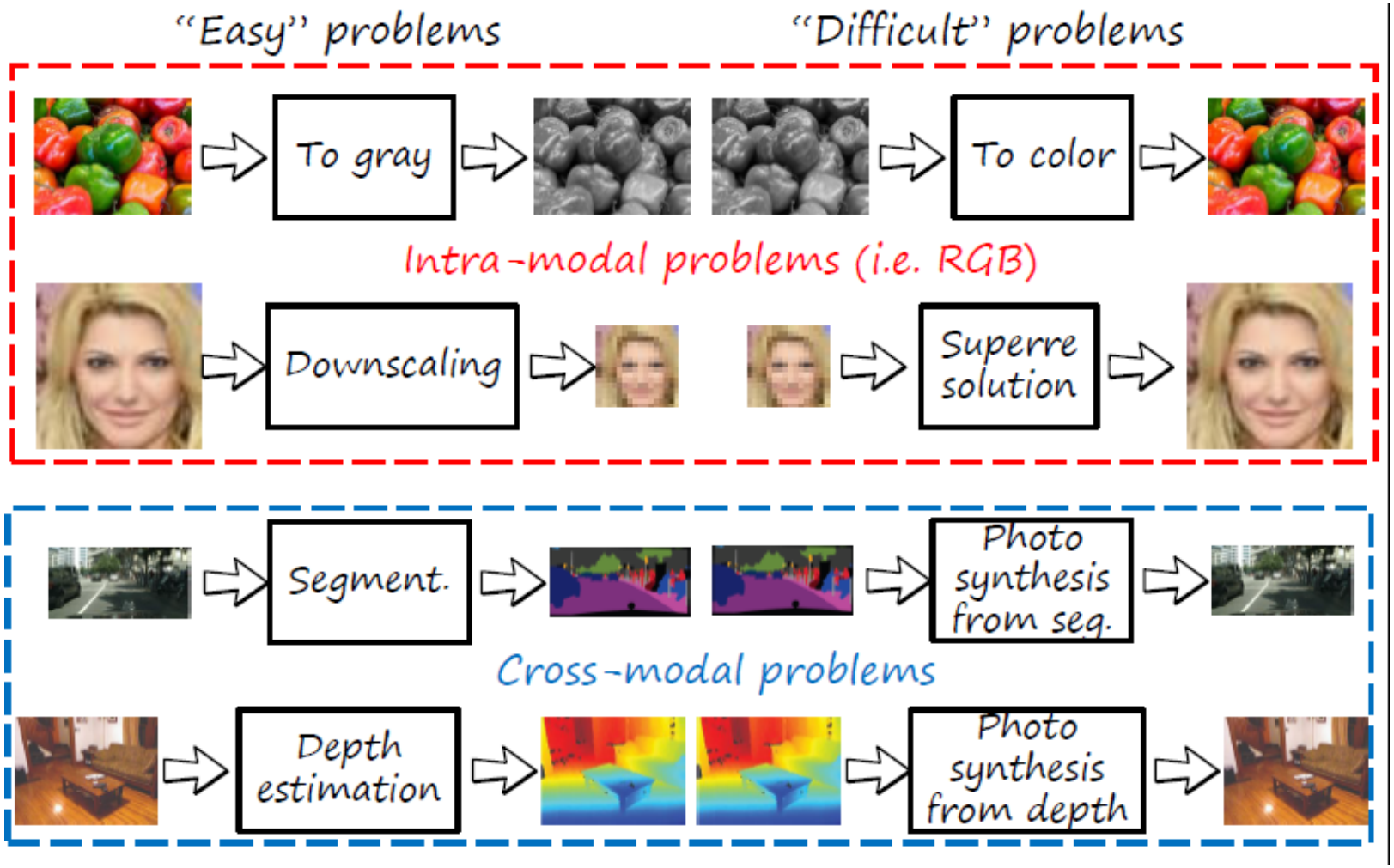


Figure 2 in the original paper.

Image-to-image translation

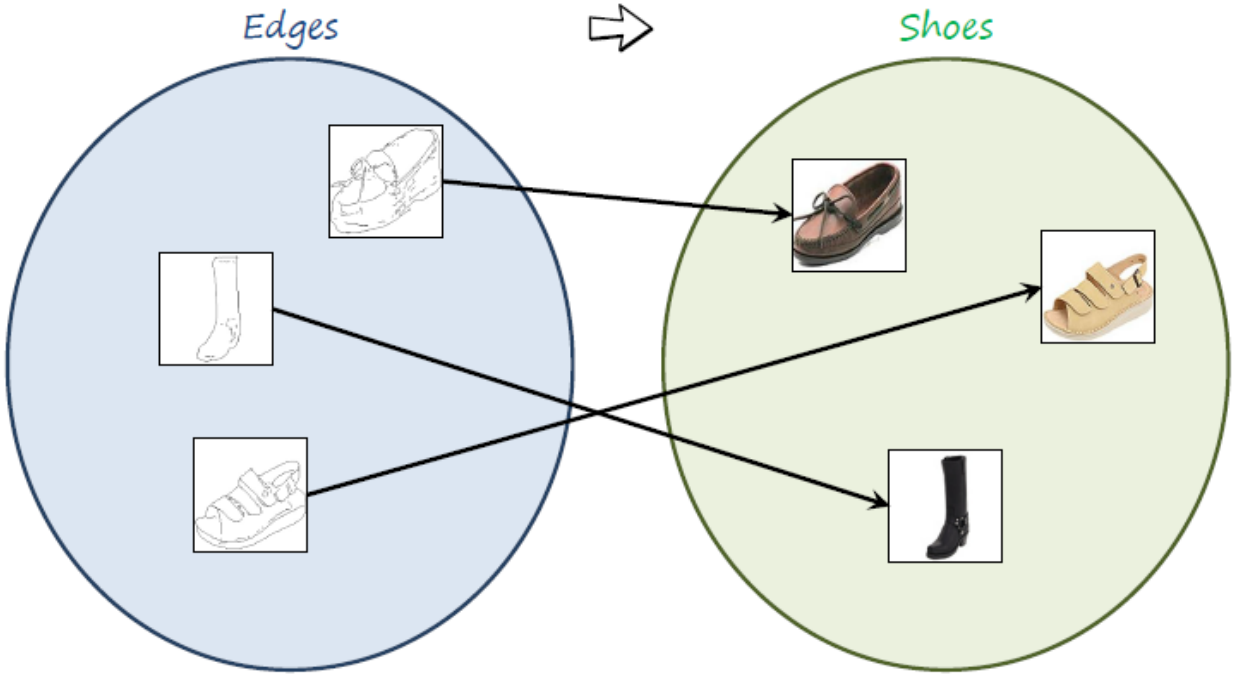


Paired Image to Image Translation

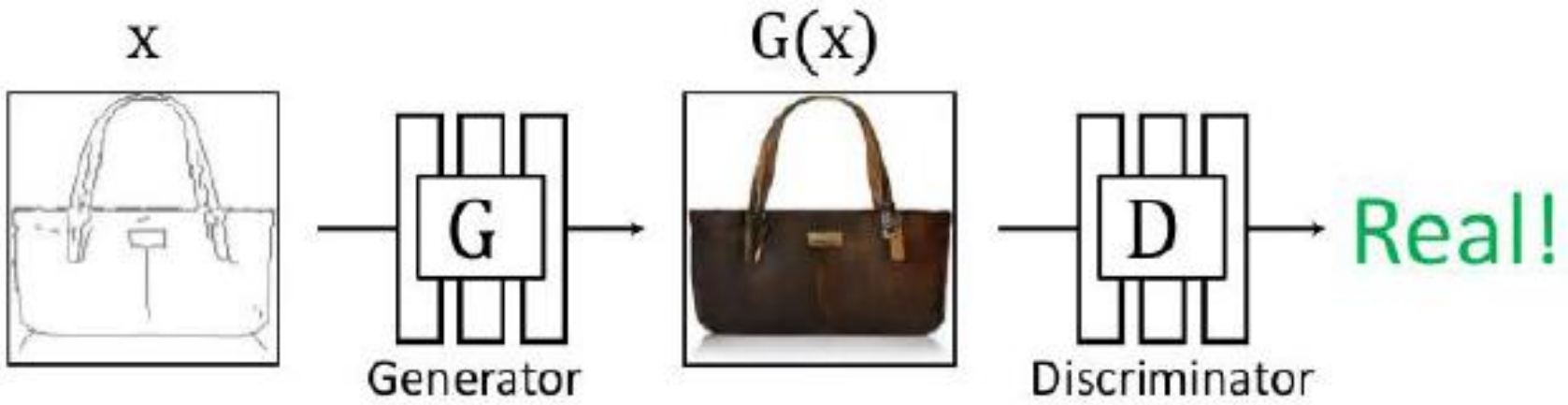


Paired Image to Image Translation

- One to One Mapping



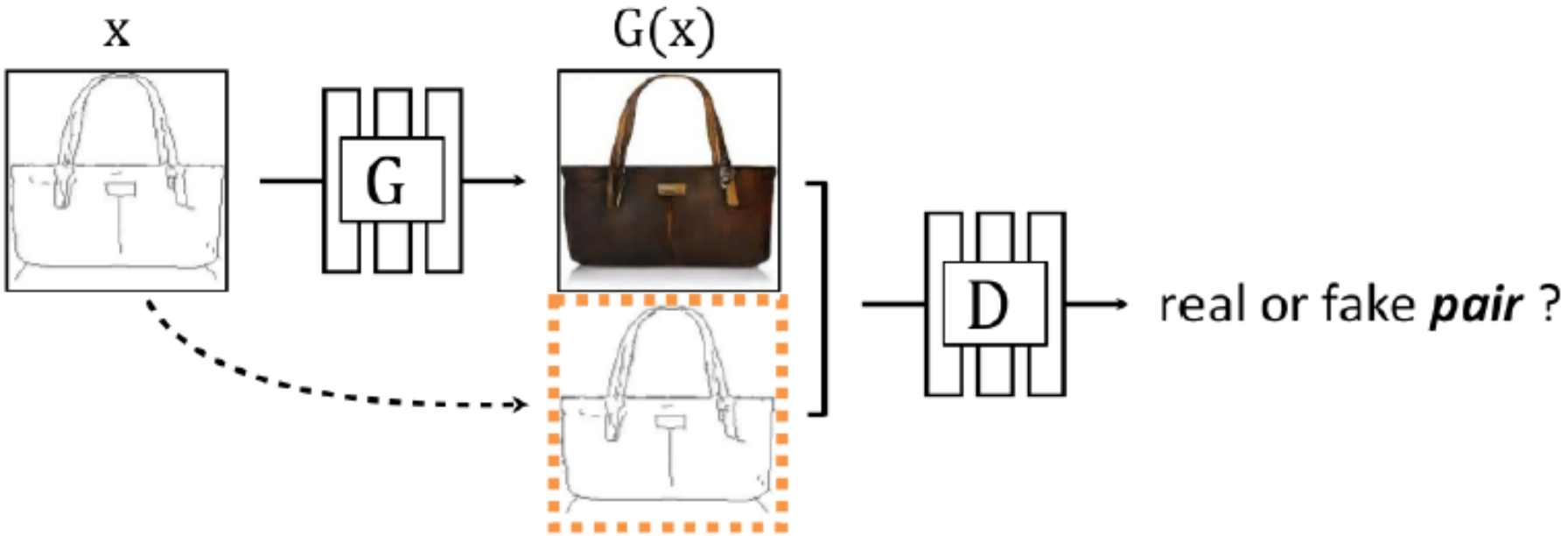
Is this conditional GAN?



$$\min_G \max_D \mathbb{E}_{x,y}[\log D(G(x)) + \log(1 - D(y))]$$

This is conventional GAN!

Pix2Pix

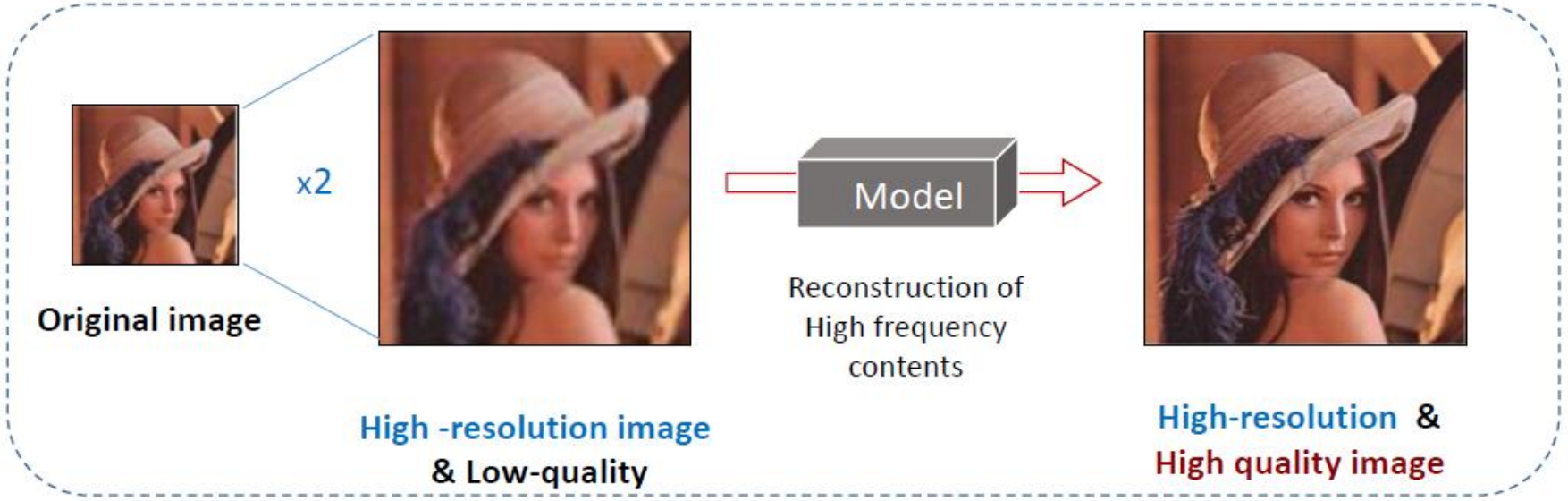


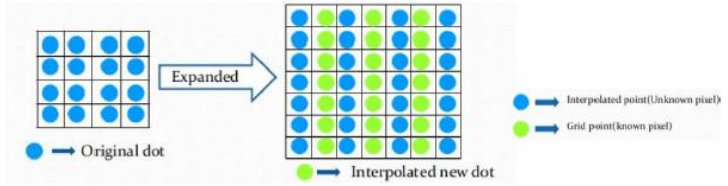
$$\min_G \max_D \mathbb{E}_{x,y}[\log D(x, G(x)) + \log(1 - D(x, y))]$$

Isola et al. "**Image-to-image translation with conditional adversarial networks**". CVPR 2017

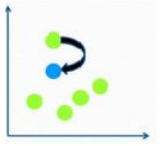
[Link to an interactive demo of this paper]

Super Resolution

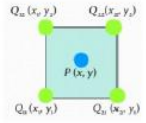




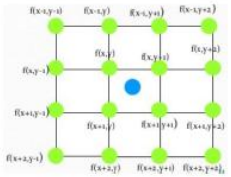
Nearest Neighbor



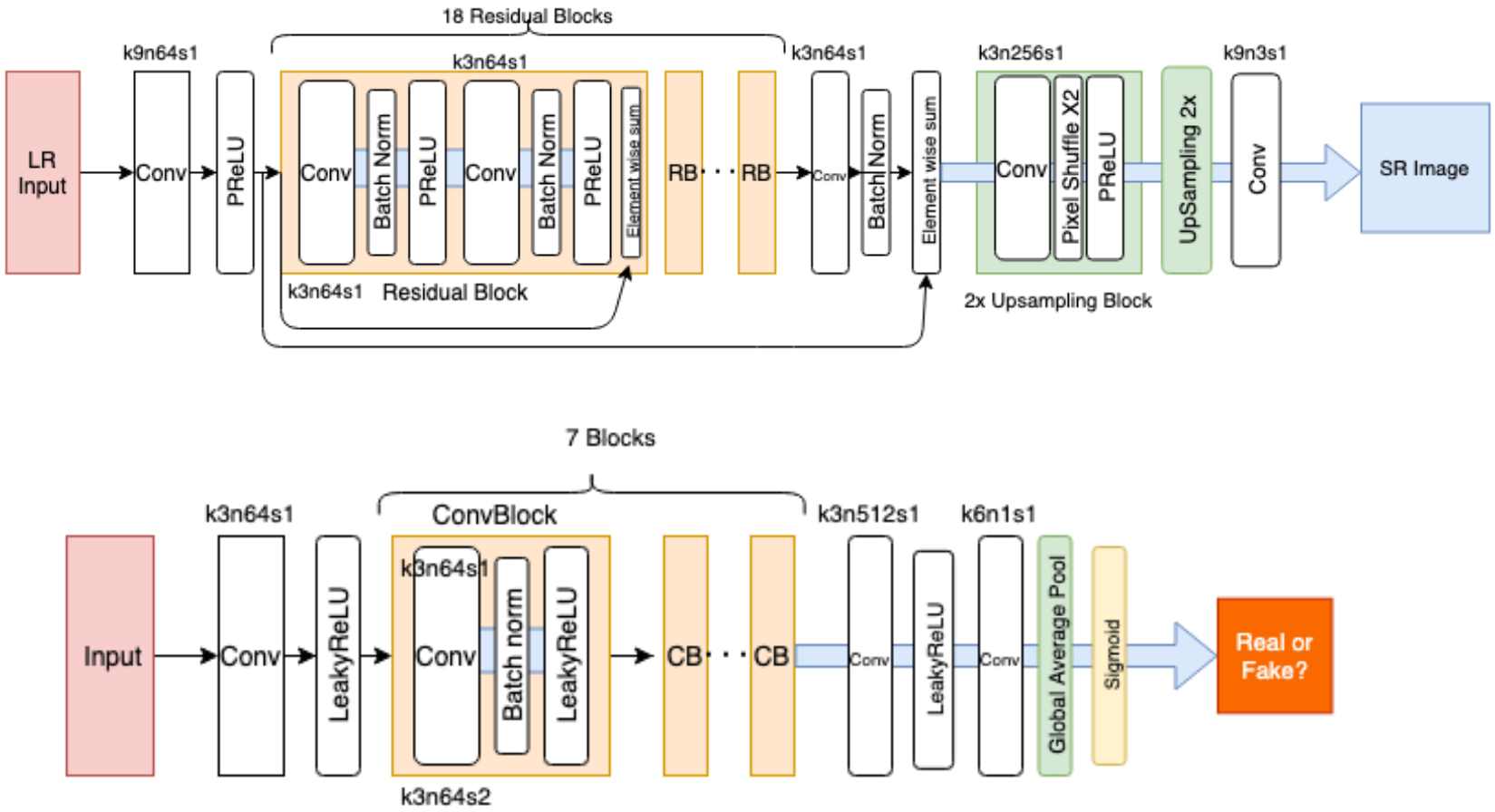
Bilinear



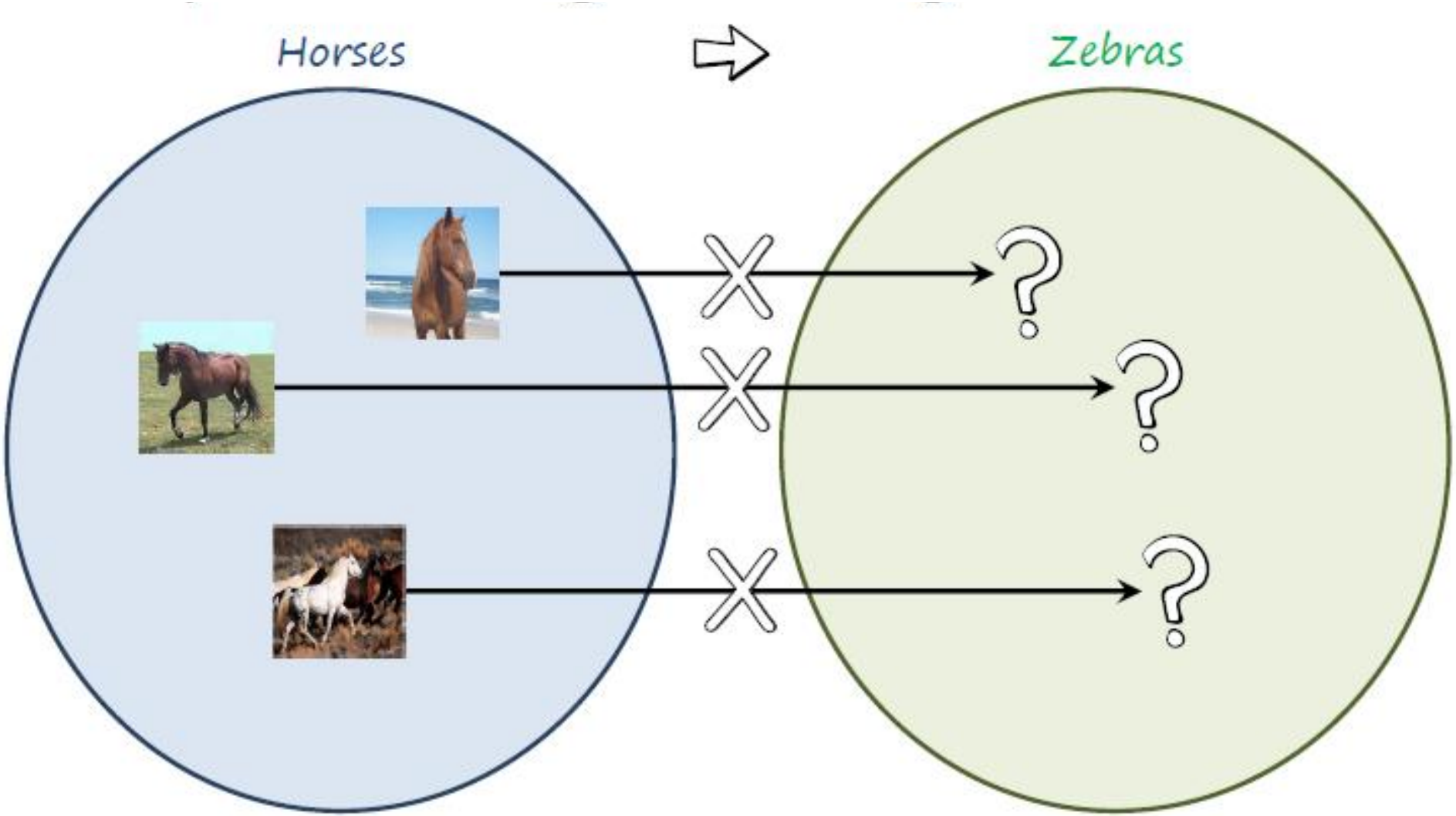
Bicubic



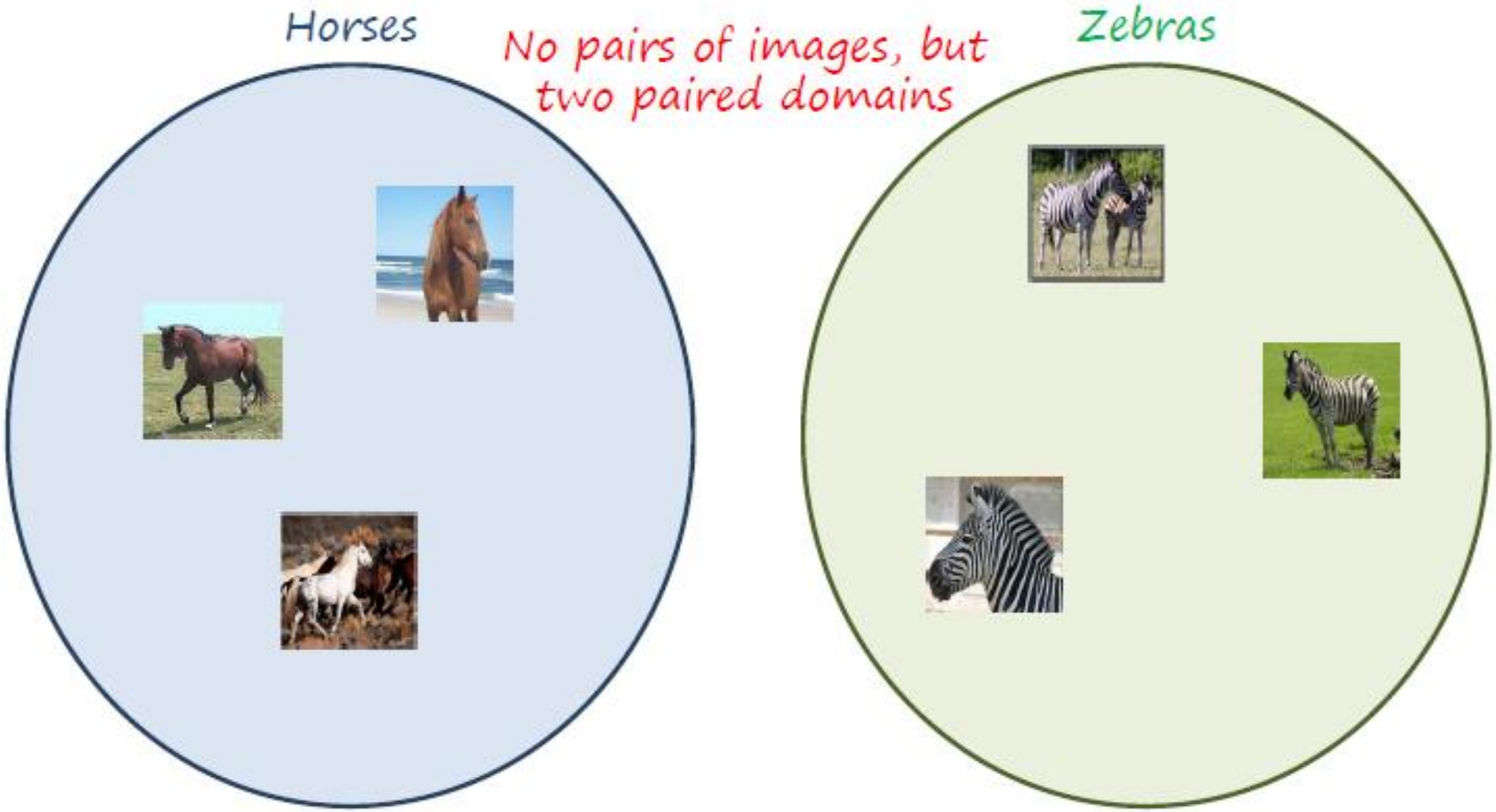
Super Resolution GAN



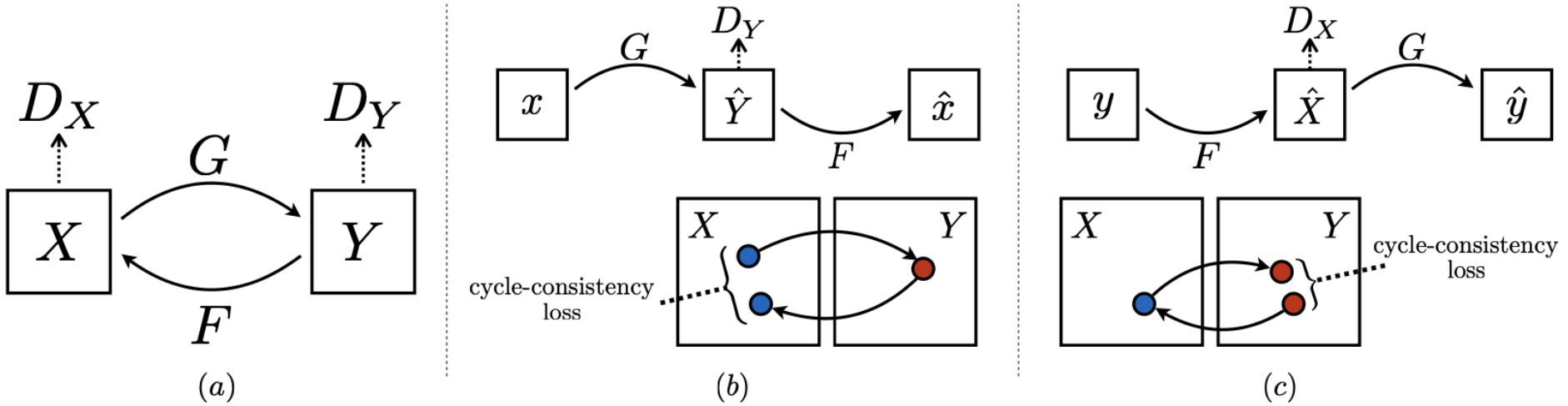
Unpaired Image to Image Translation



Unpaired Image to Image Translation



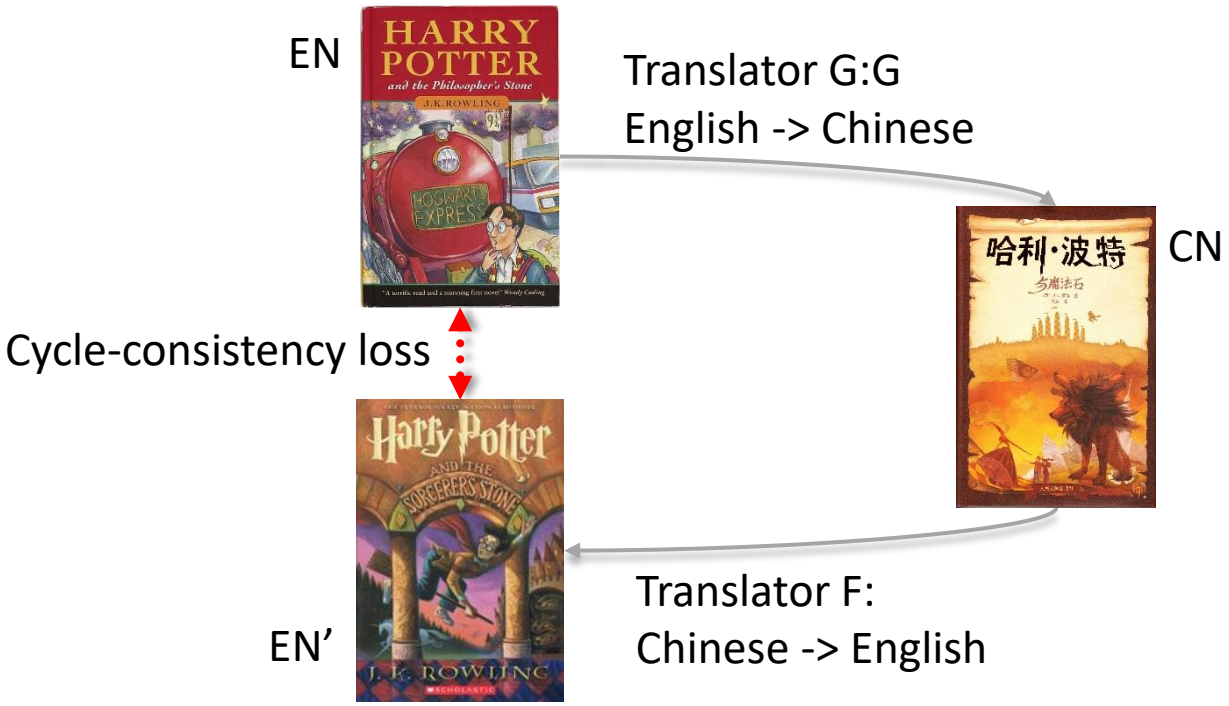
CycleGAN



The model contains two mapping functions G : X → Y and F : Y → X, and associated adversarial discriminators. It has two cycle consistency losses that capture the intuition that if it translates from one domain to the other and back again it should arrive at where it started: (b) forward cycle-consistency loss: x → G(x) → F (G(x)) ≈ x, and (c) backward cycle-consistency loss: y → F (y) → G(F (y)) ≈ y

Zhu et al., "Unpaired Image-to-Image Translation Using Cycle-Consistent Adversarial Networks," *2017 IEEE International Conference on Computer Vision (ICCV)*, pp. 2242-2251.

Cycle Consistency

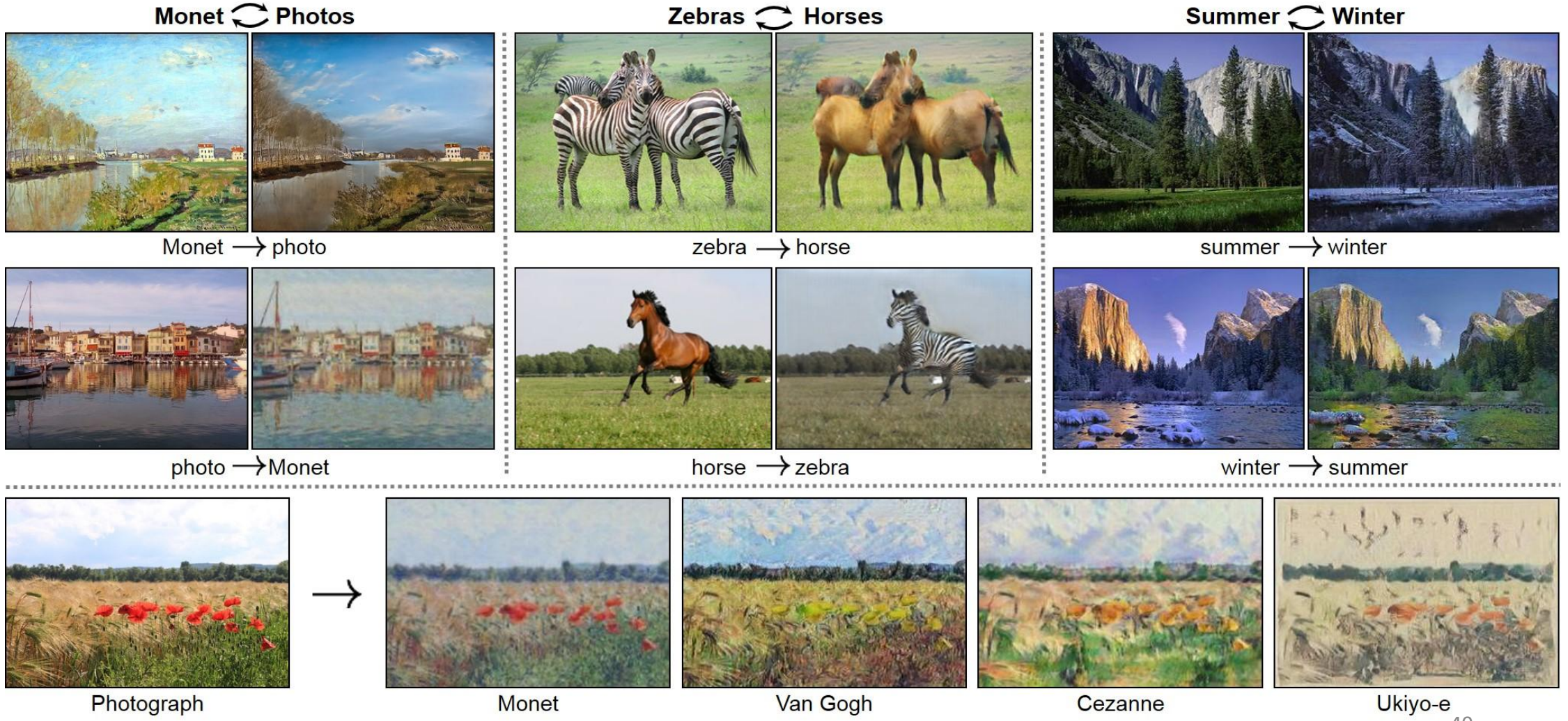


Obviously, we expected

$$EN \rightarrow \underbrace{G(EN)}_{CN} \rightarrow \underbrace{F(G(EN))}_{EN'} \sim EN$$

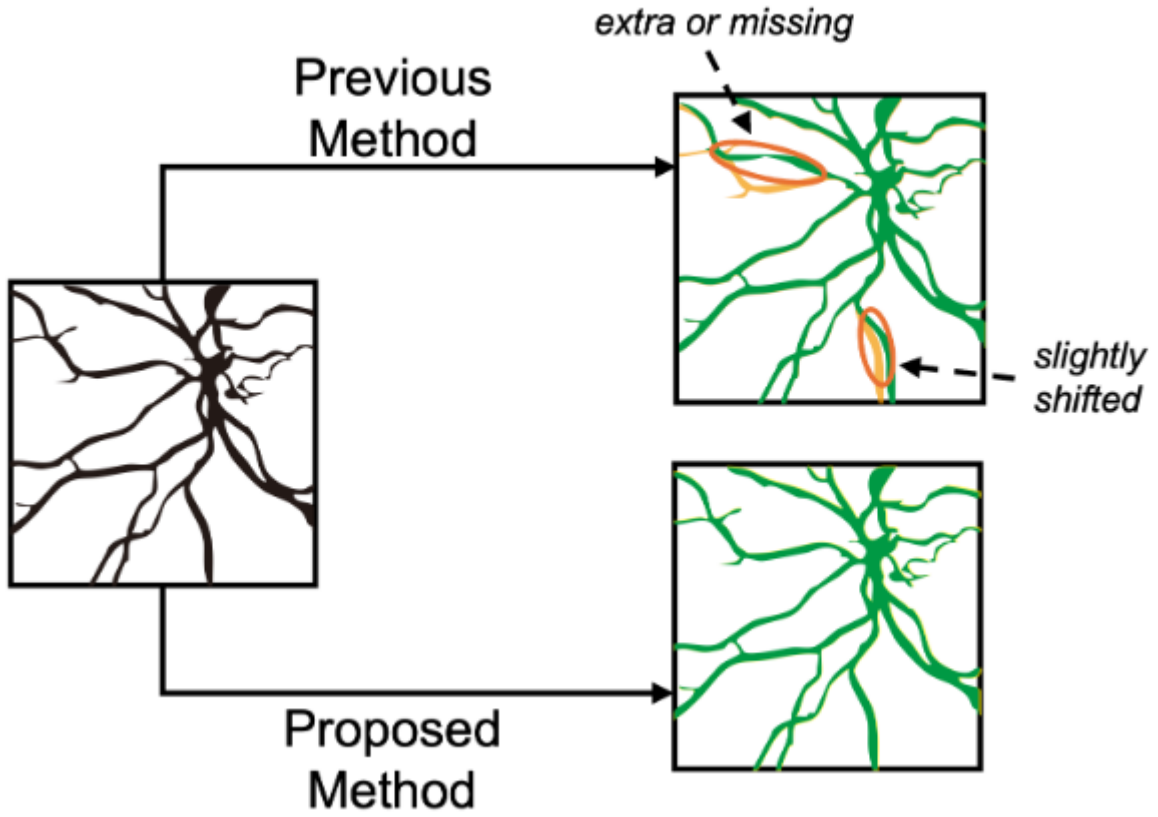
Formalized,
if G: X -> Y, F: Y -> X; expect,
X -> G(X) -> F(G(X)) ~ X

CycleGAN

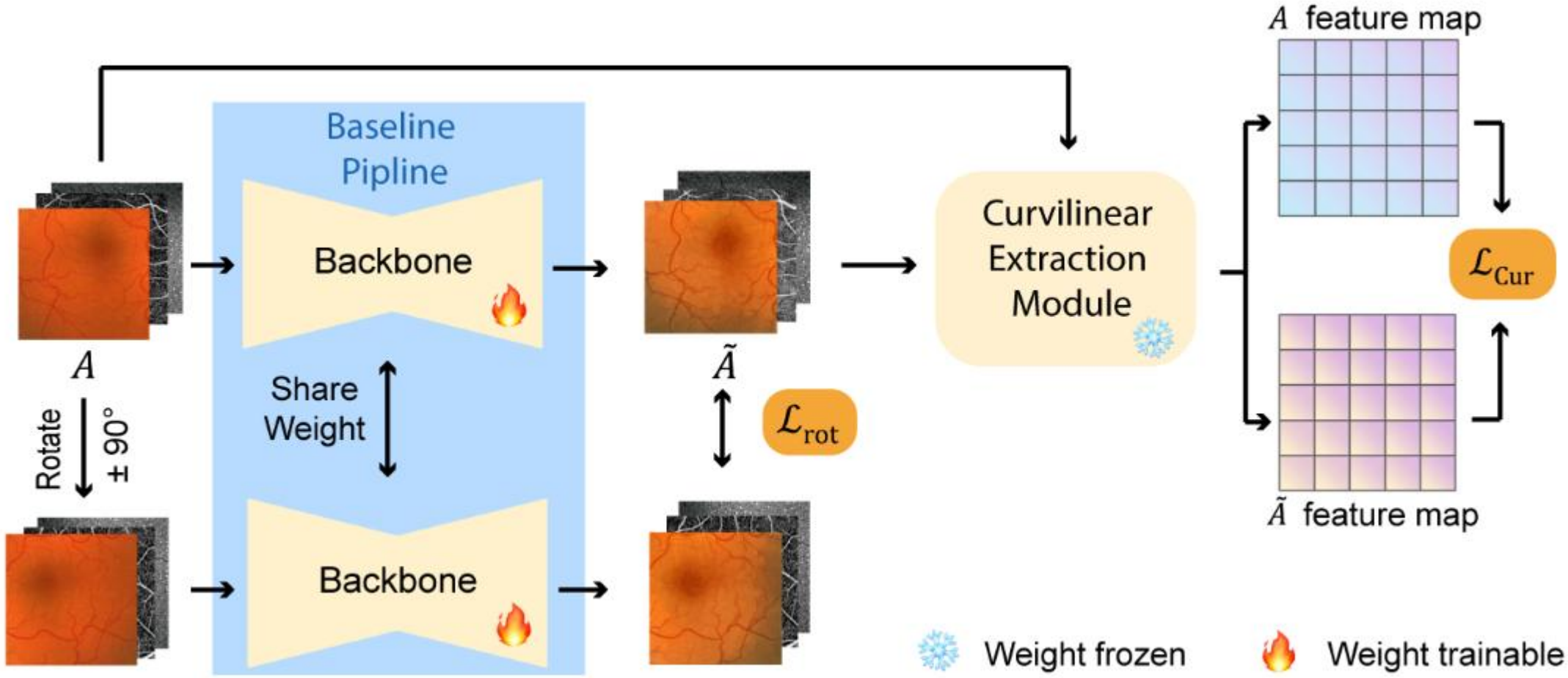

Monet ⇄ Photos
Zebras ⇄ Horses
Summer ⇄ Winter
Monet → photo
zebra → horse
summer → winter
photo → Monet
horse → zebra
winter → summer
Photograph
Monet
Van Gogh
Cezanne
Ukiyo-e

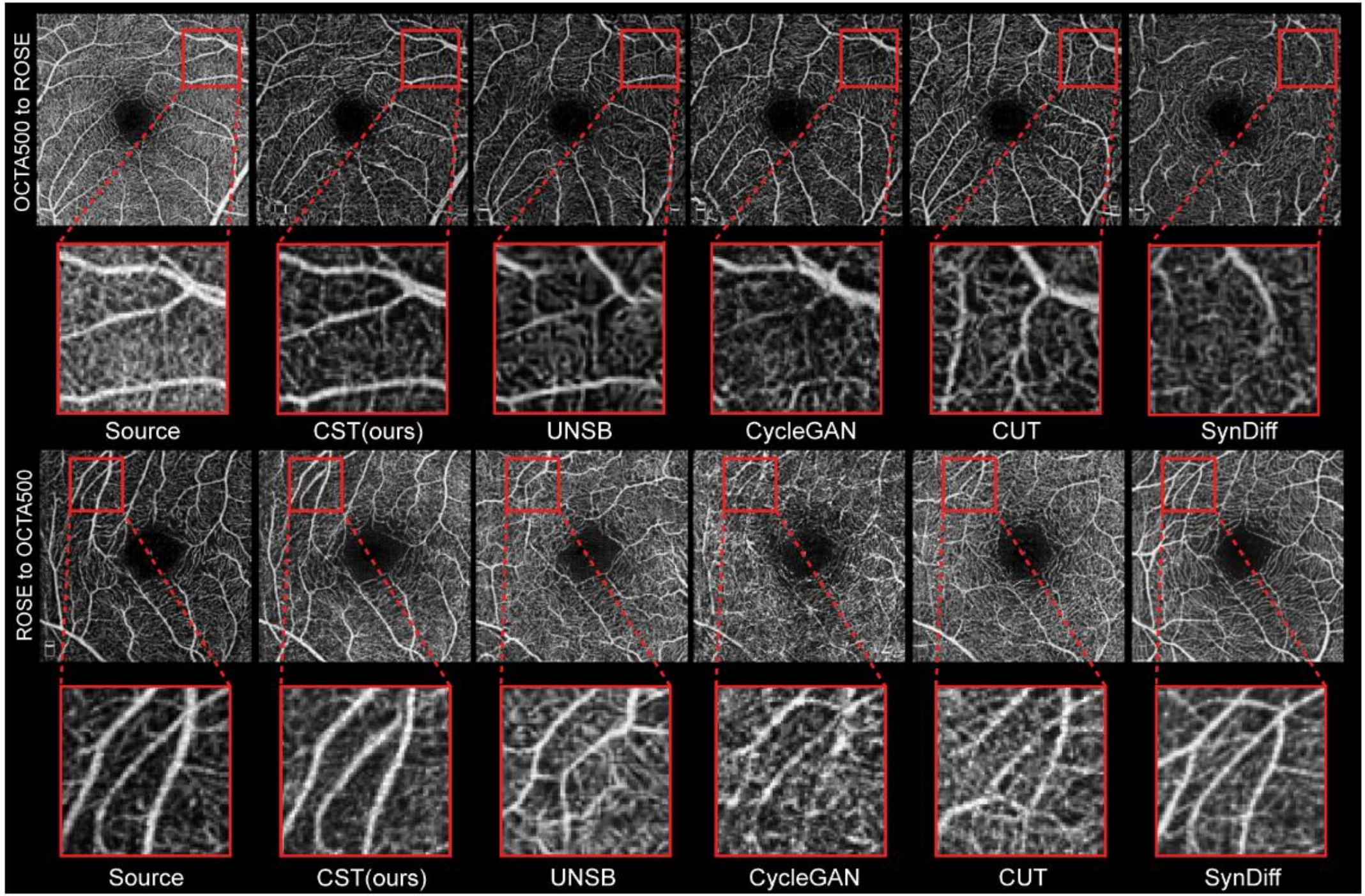
Structure-Preserving Translation

Motivation: structure inconsistency

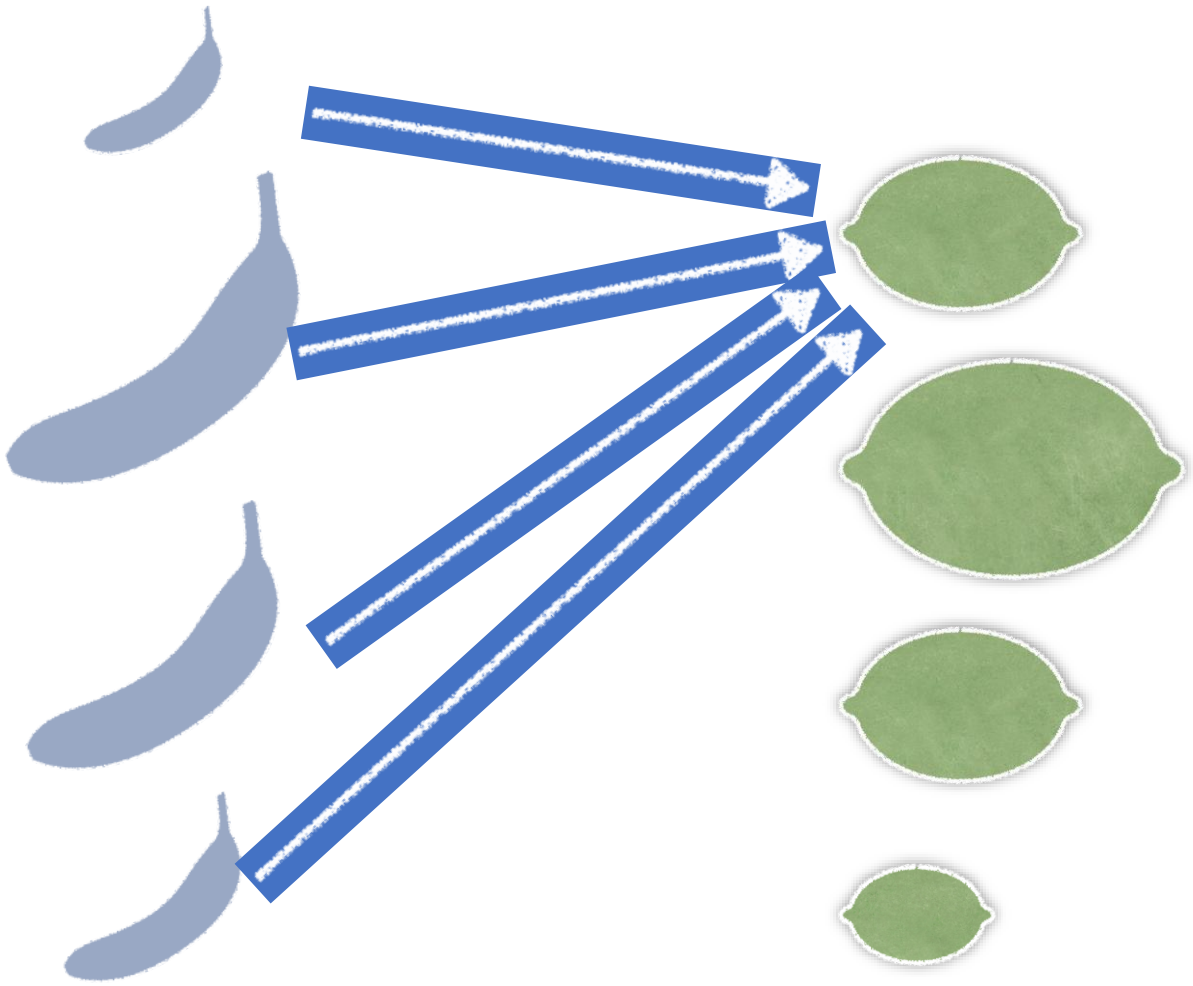


Structure-Preserving Translation

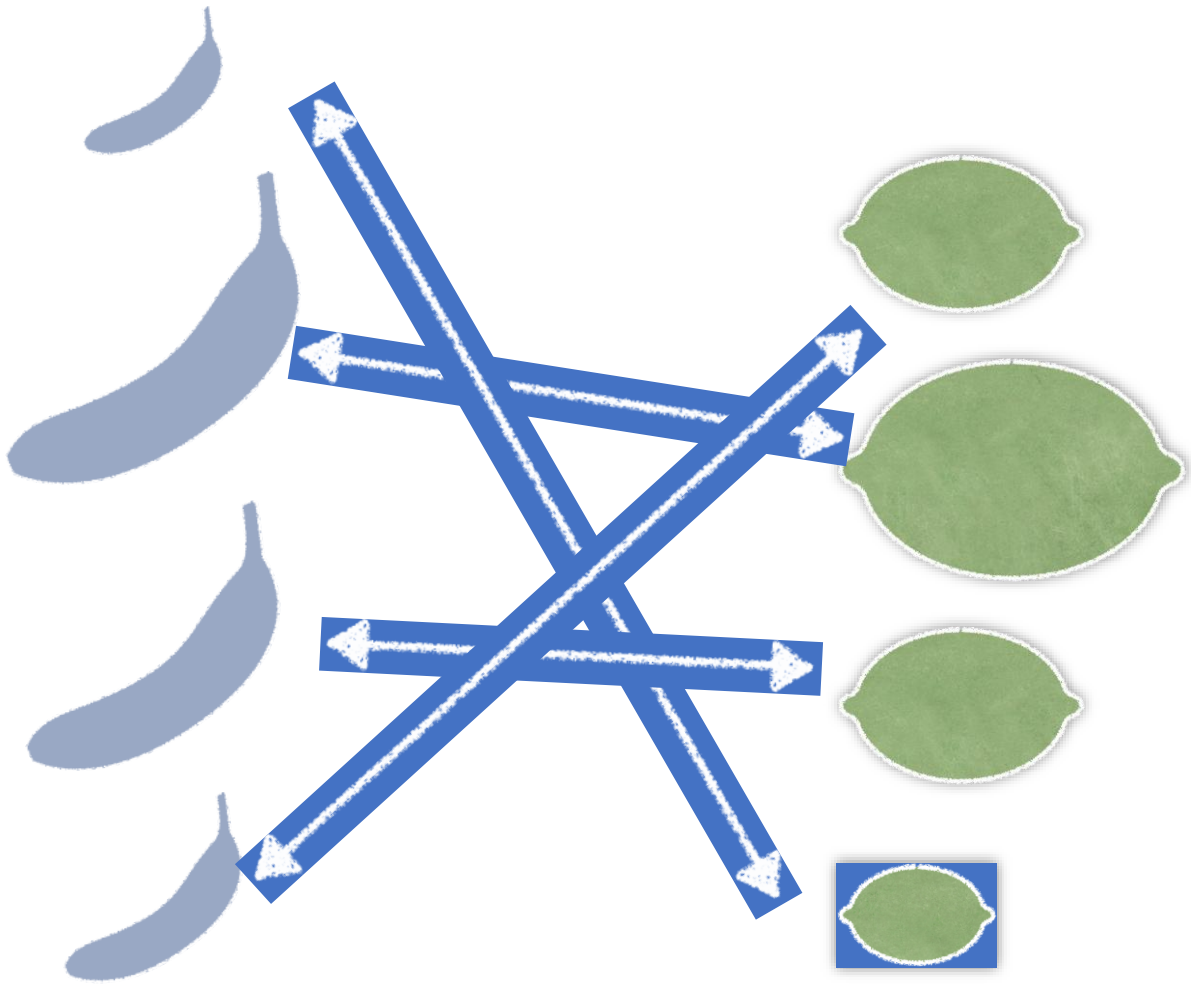



OCTA500 to ROSE
Source
CST(ours)
UNSB
CycleGAN
CUT
SynDiff
ROSE to OCTA500
Source
CST(ours)
UNSB
CycleGAN
CUT
SynDiff

mode collapse



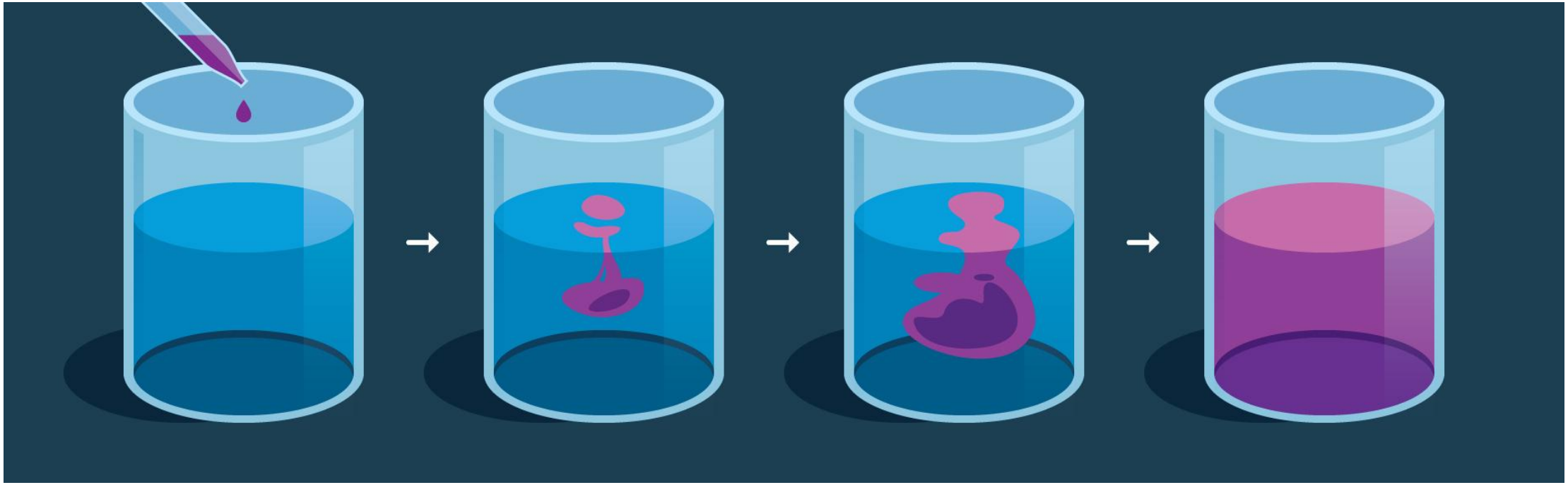
one to one mapping



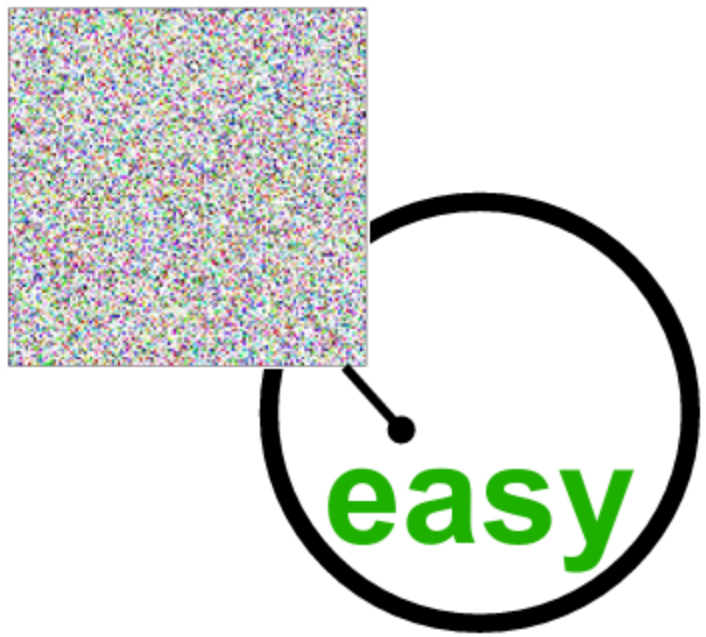
Diffusion based Video Generation

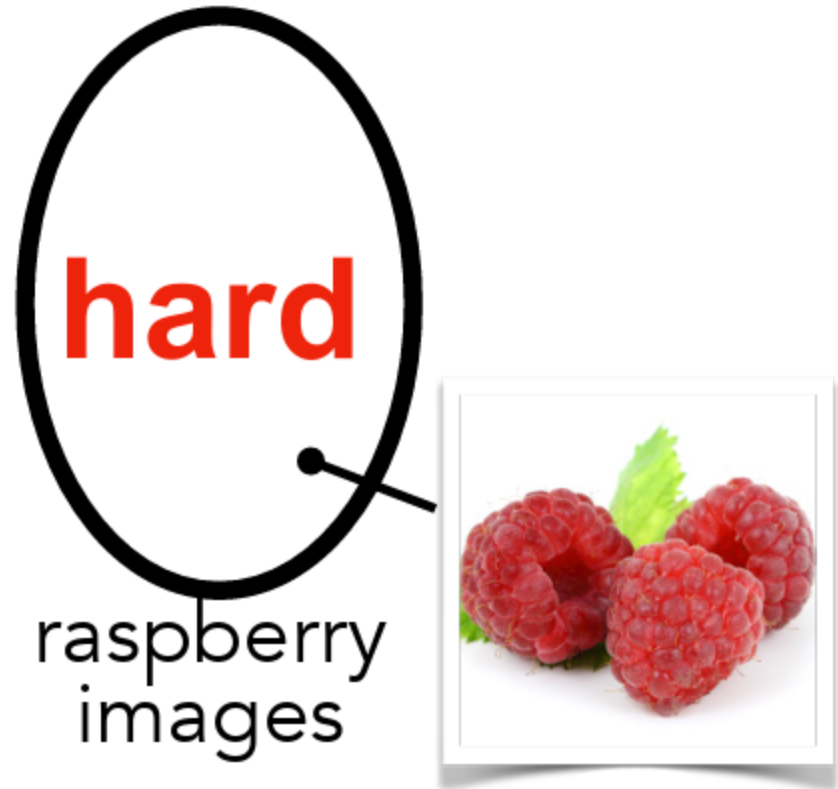
What is diffusion

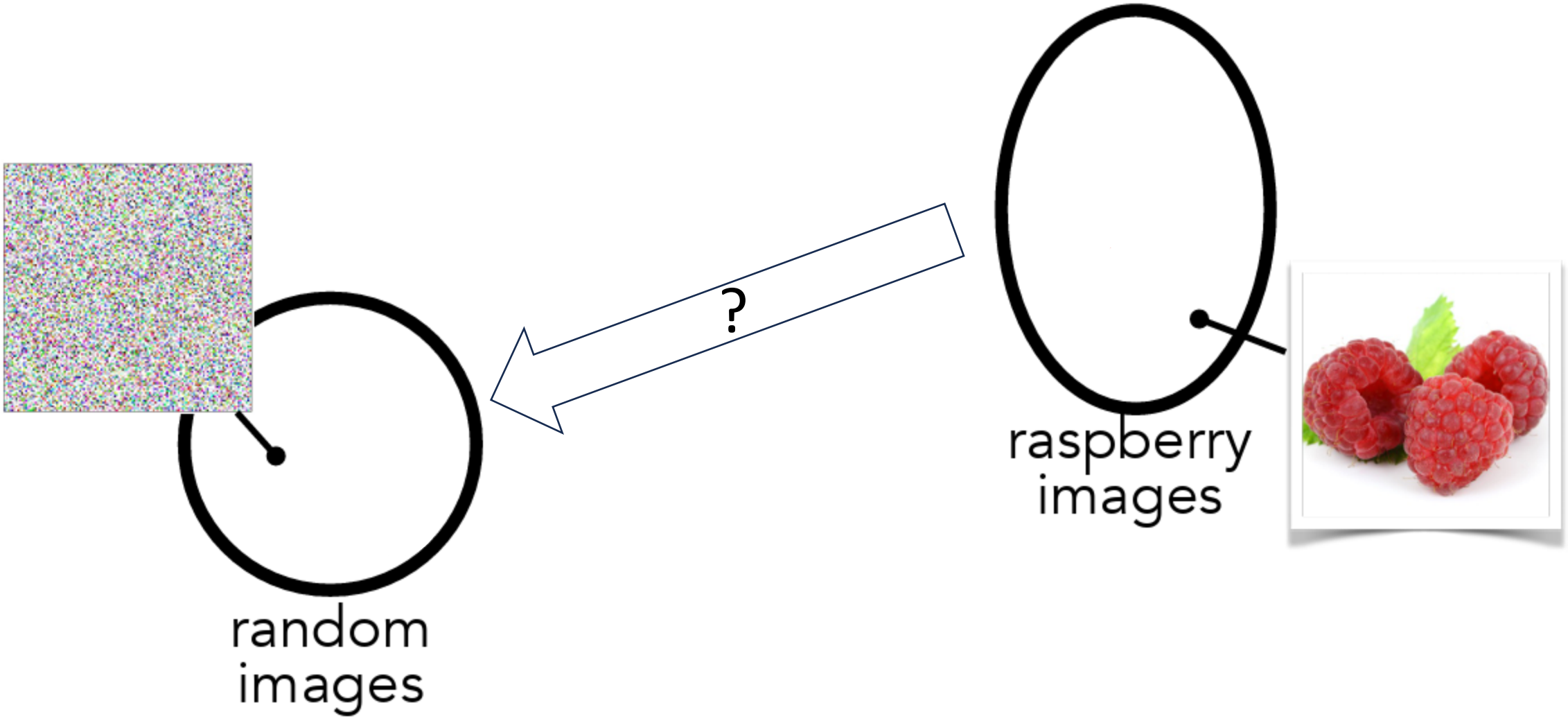
In Science, Diffusion is a process where solute particles move from higher concentrated region of solute to lower concentrated region of solute in a solvent. Consider the below diffusion analogy for high-level intuition:

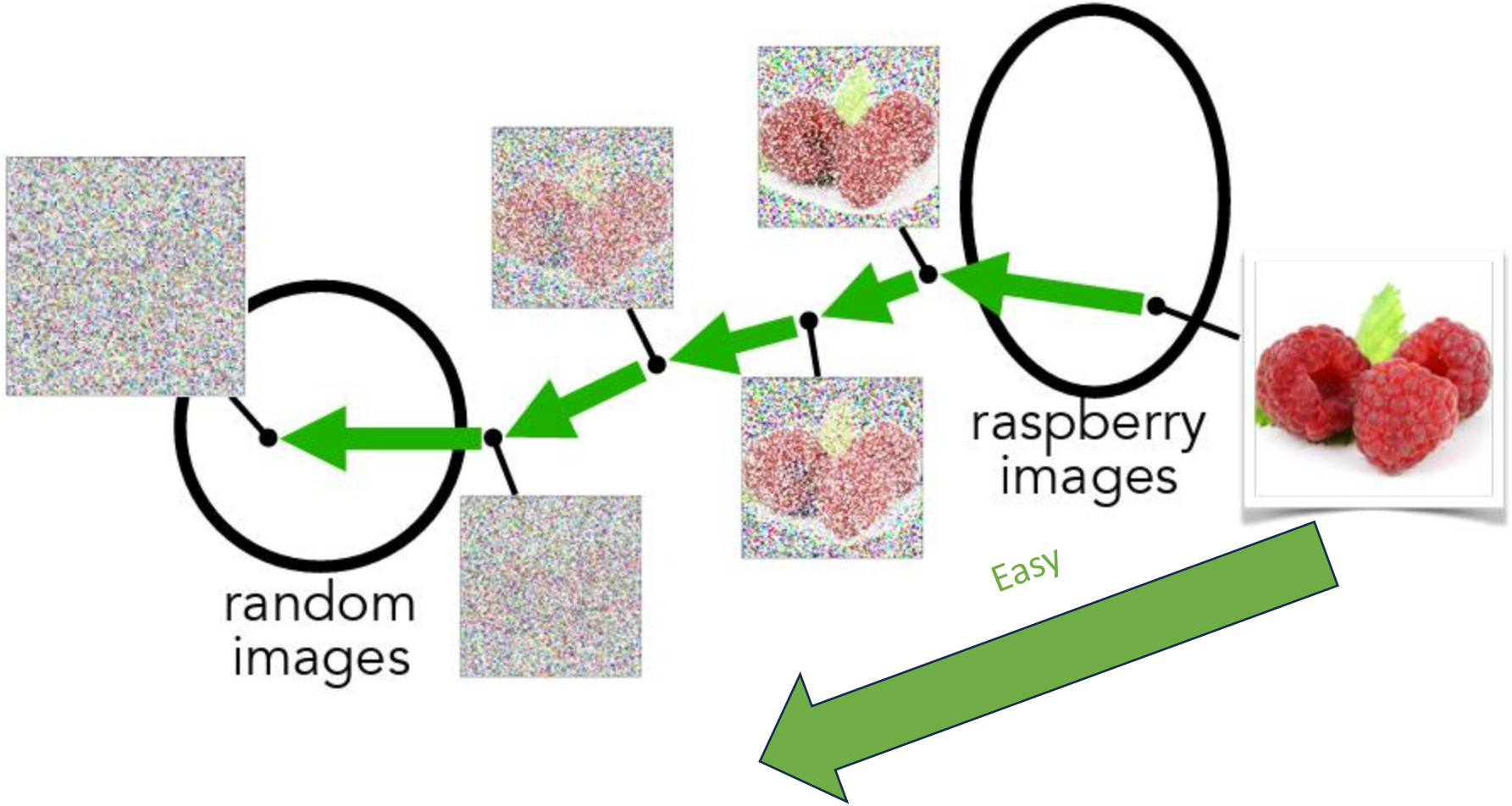


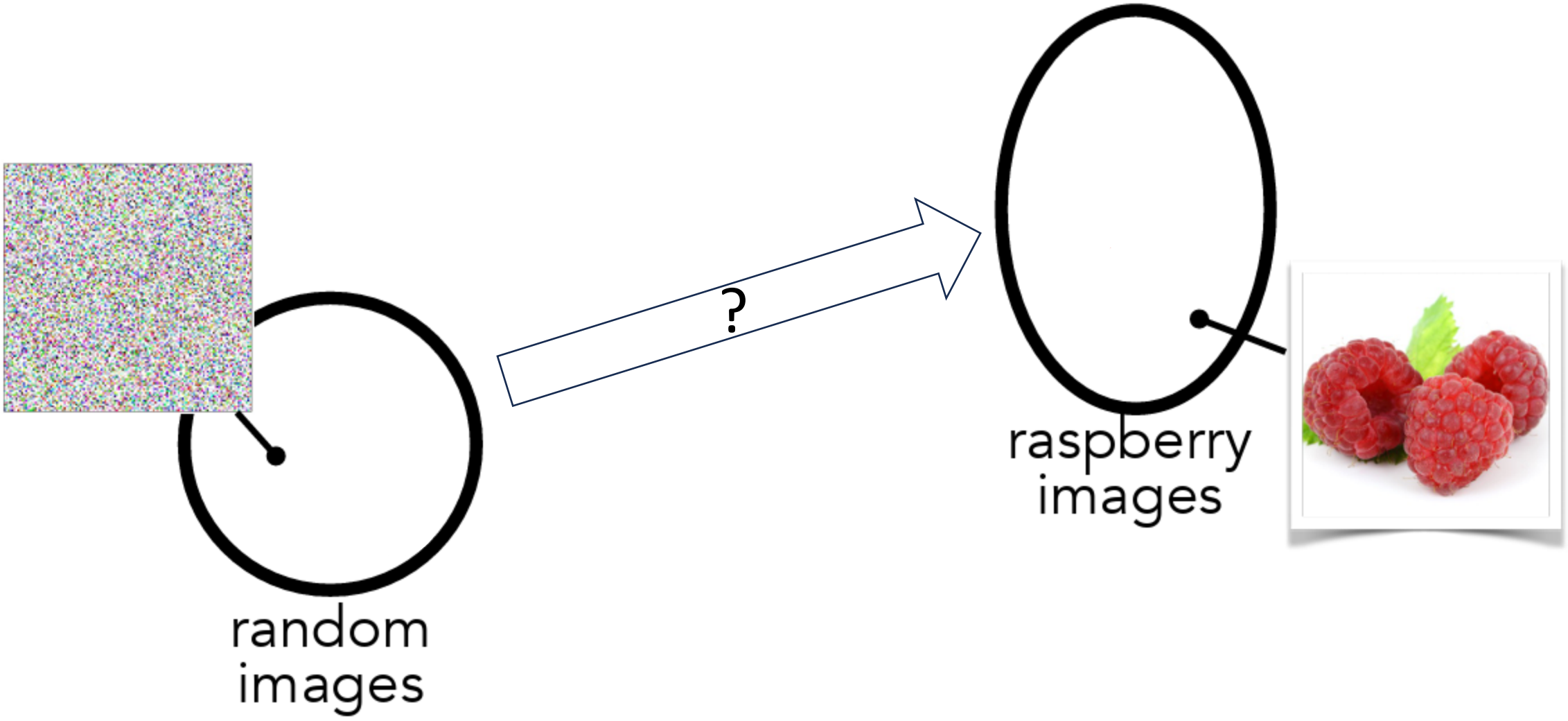
Figures from https://huggingface.co/learn/computer-vision-course/unit5/generative-models/diffusion-models/introduction

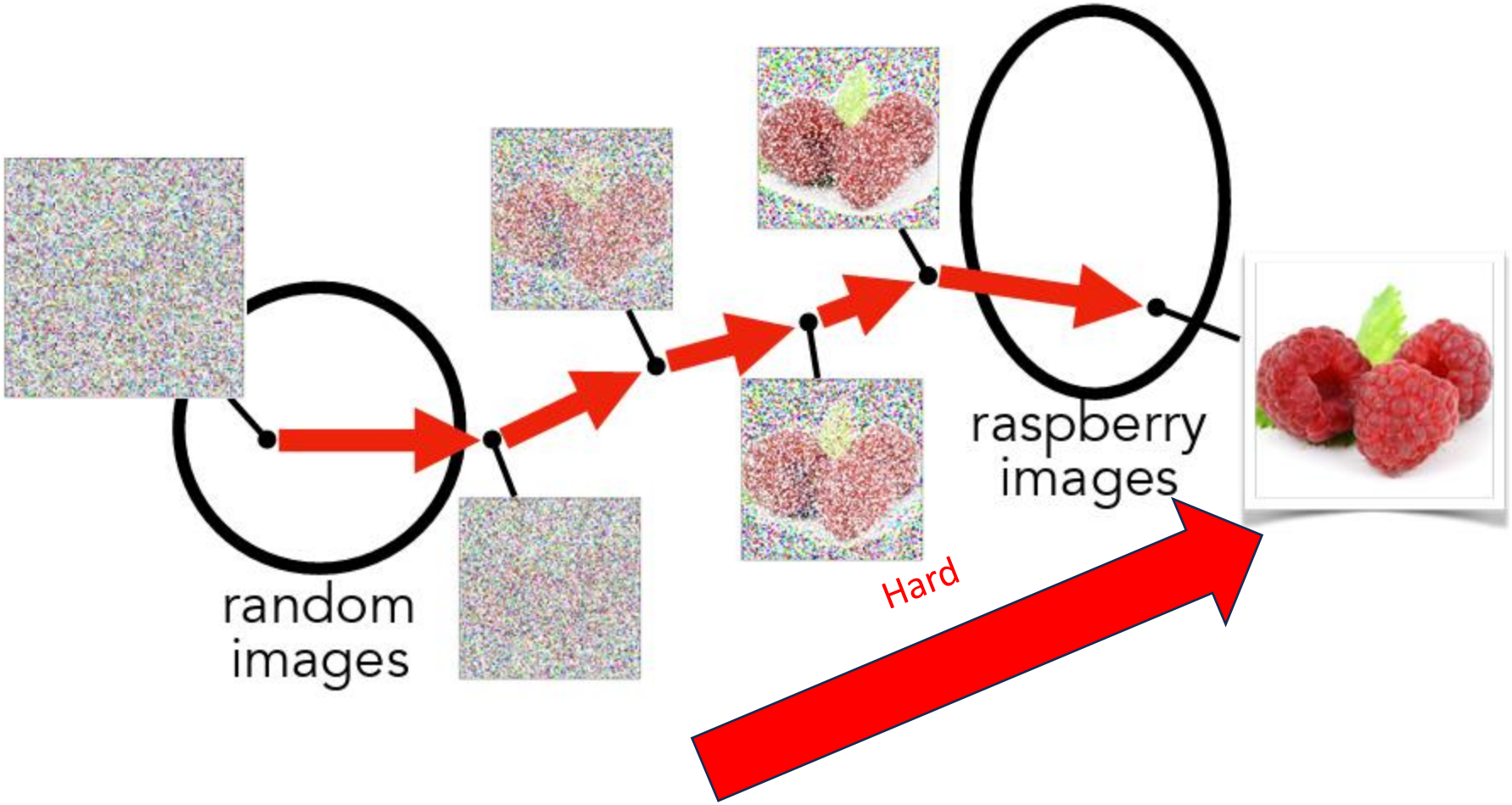

easy
random
images

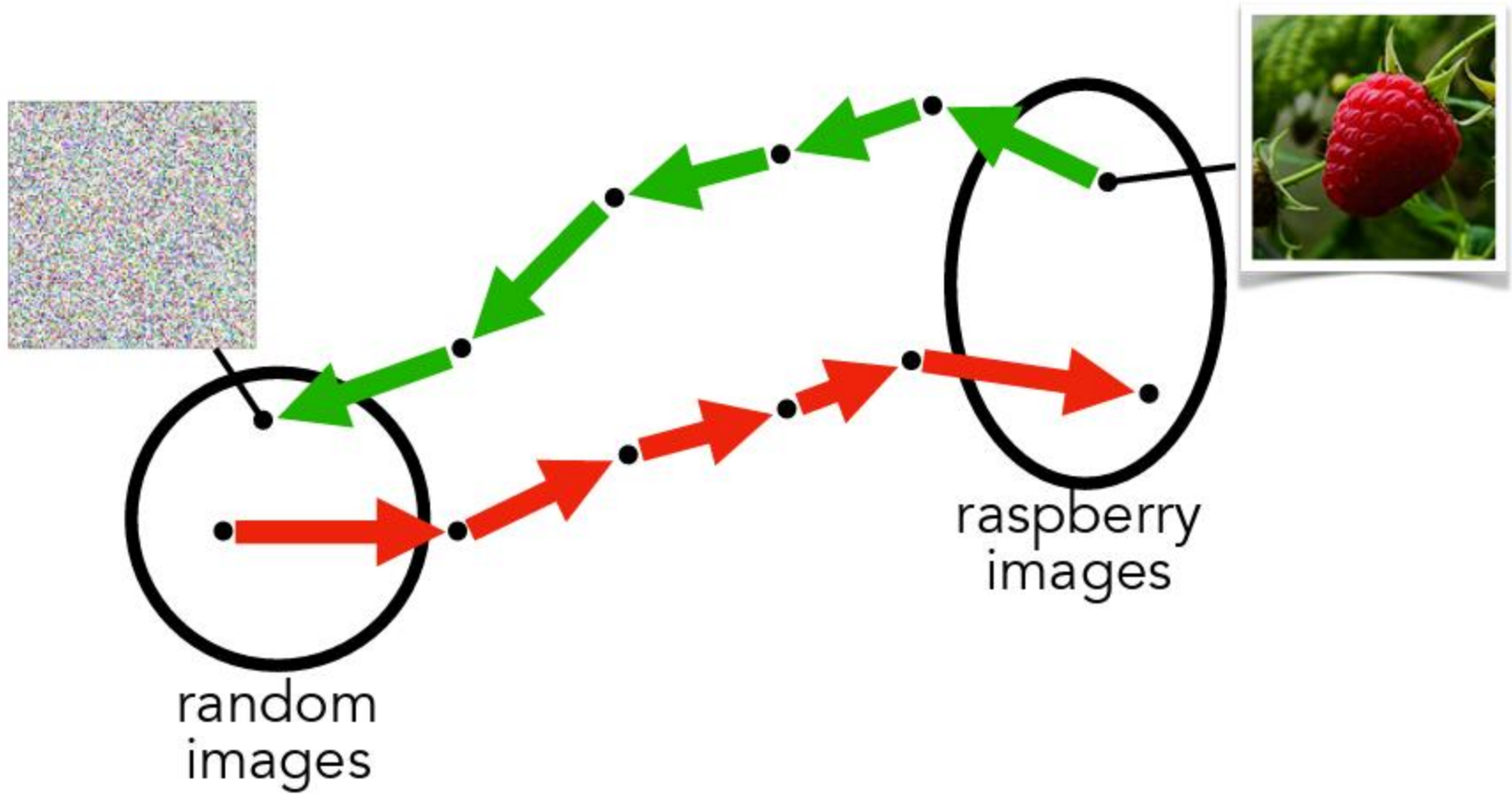

hard
raspberry
images

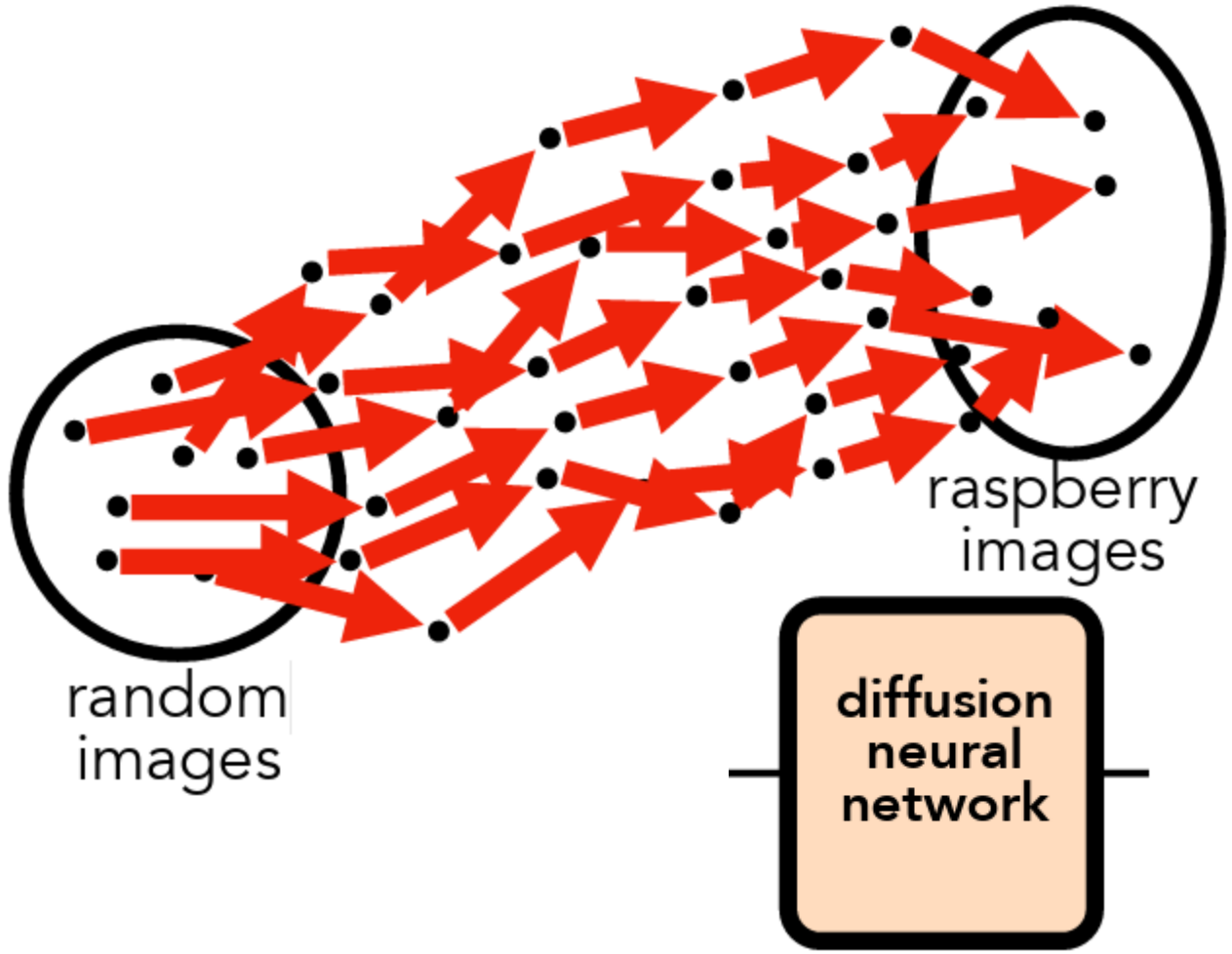

?
raspberry
images
random
images

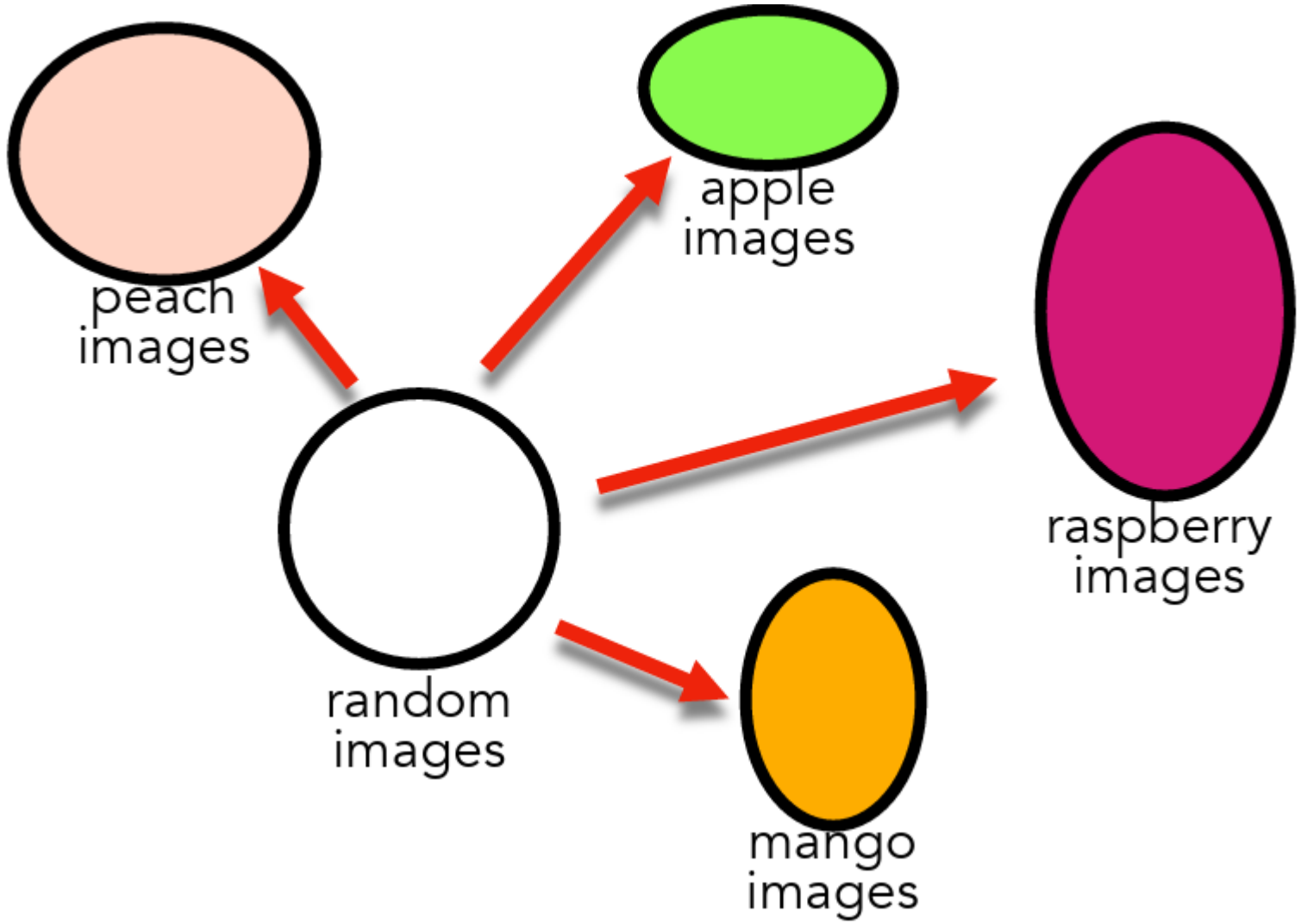

raspberry
images
random
images
Easy

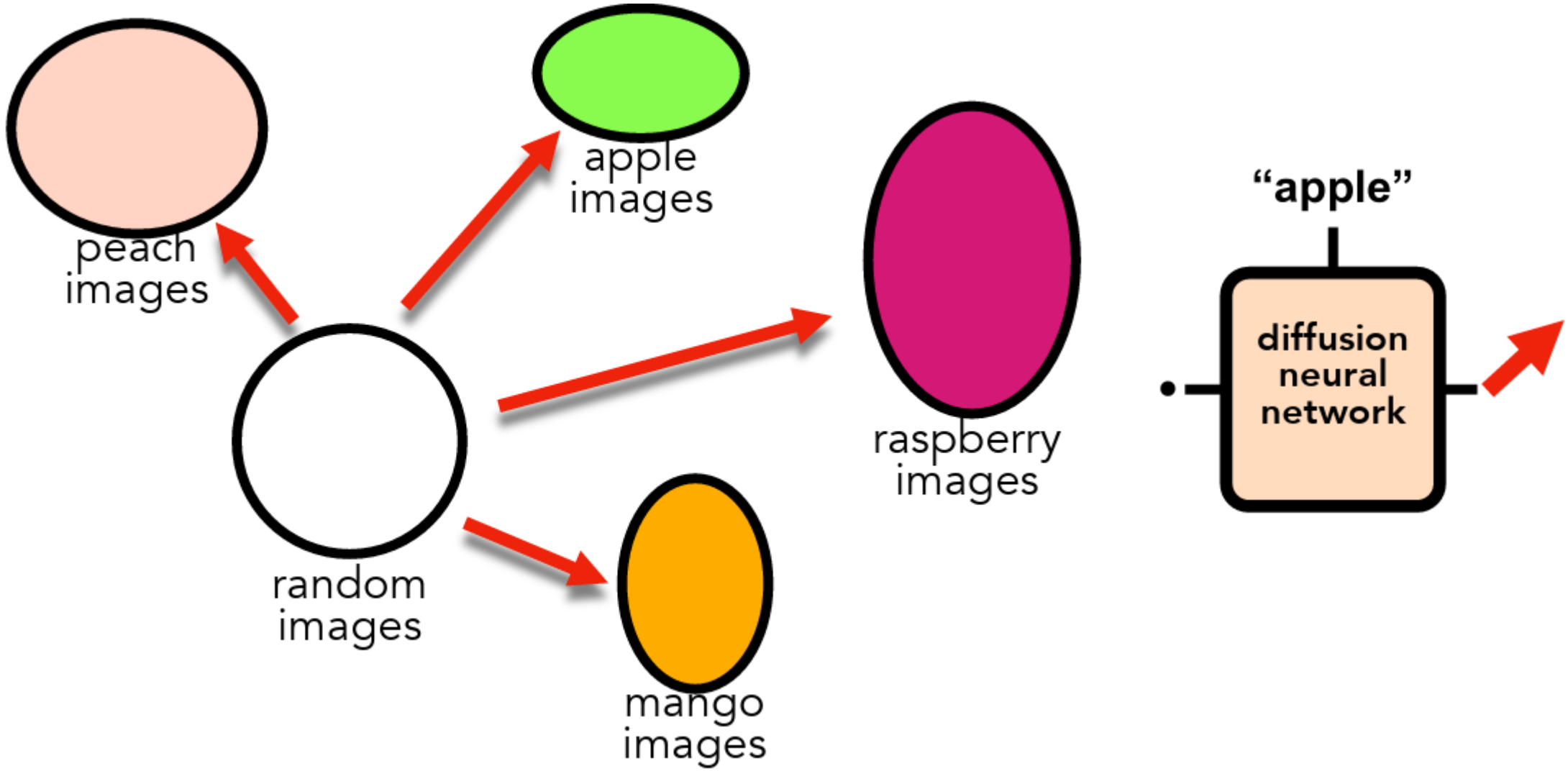

?
raspberry
images
random
images

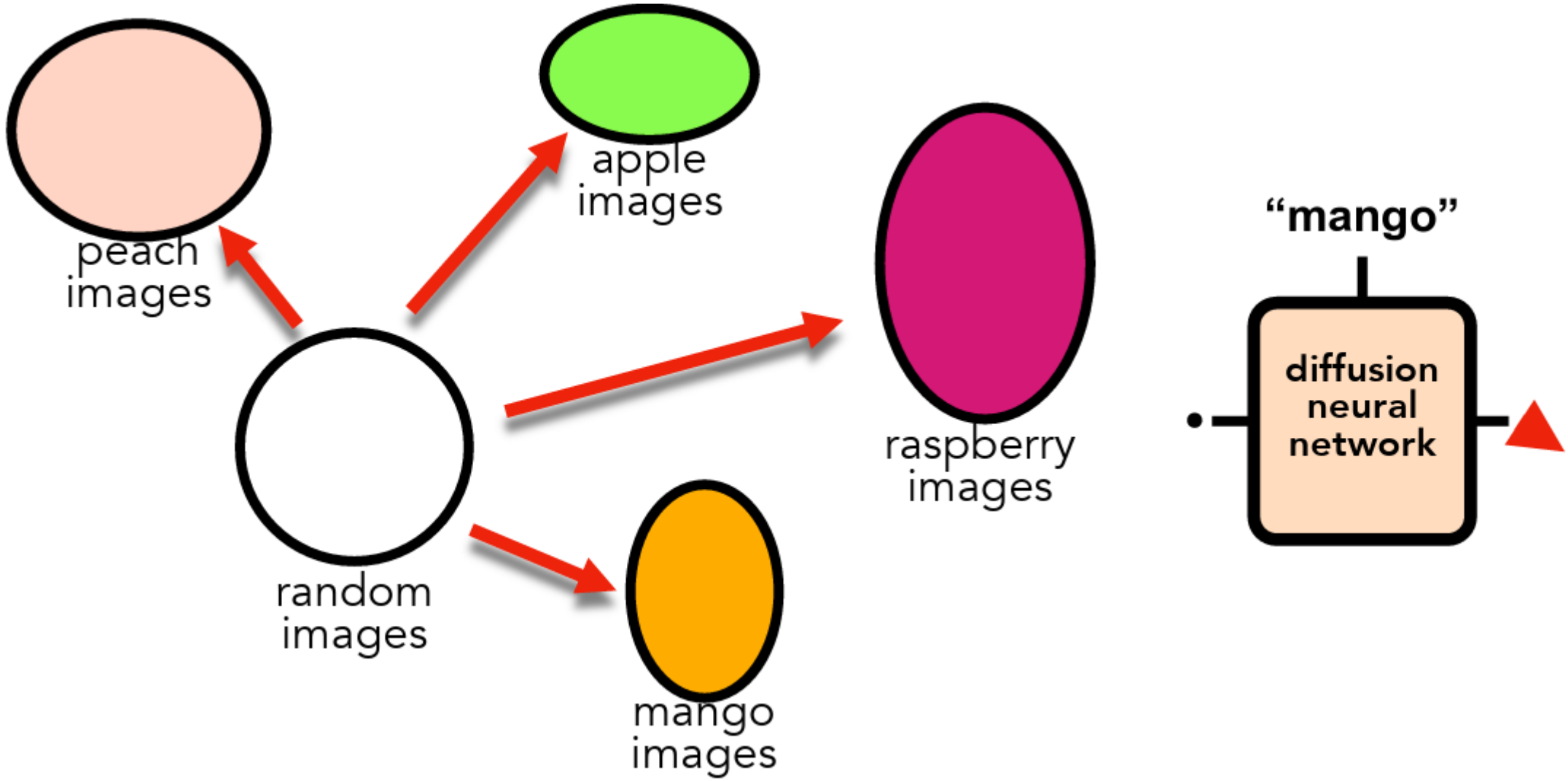

random
images
raspberry
images
Hard

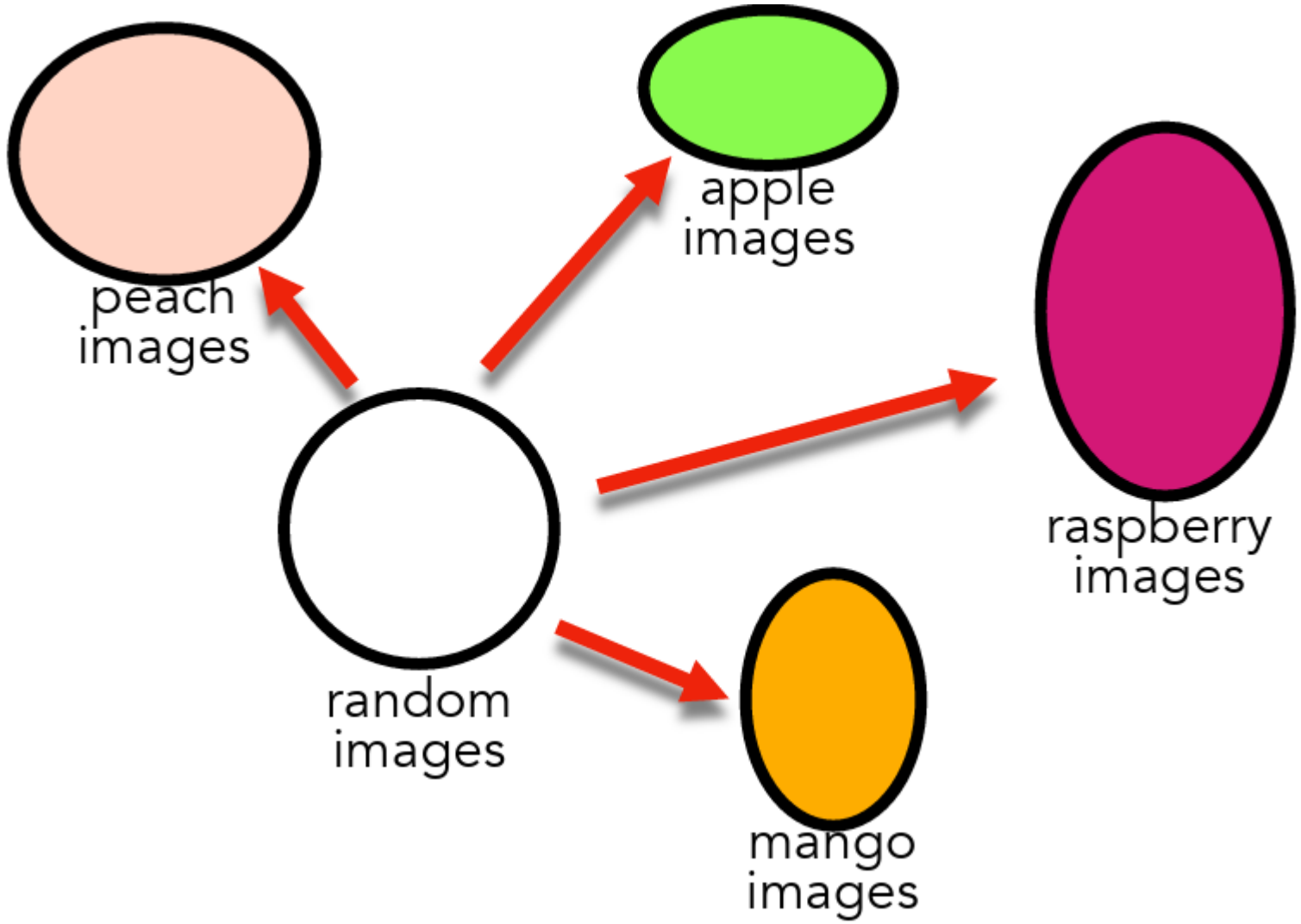

raspberry
images
random
images

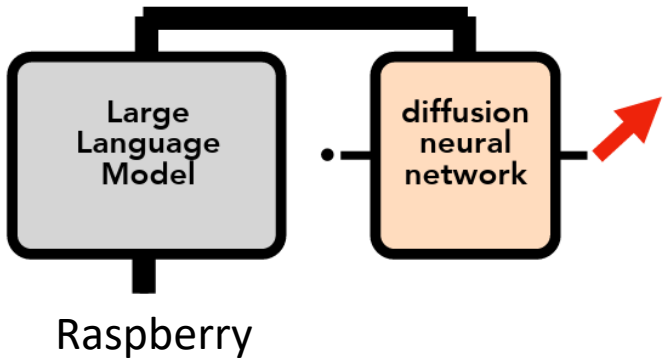

raspberry
images
random
images
diffusion
neural
network


peach
images
apple
images
raspberry
images
random
images
mango
images


peach images
apple images
random images
raspberry images
mango images
“apple”
diffusion neural network


peach
images
apple
images
raspberry
images
random
images
mango
images
“mango”
diffusion
neural
network


peach
images
apple
images
raspberry
images
random
images
mango
images


Large
Language
Model
diffusion
neural
network
Raspberry

Text to Image

OpenAI's DALL-E 1,2,3



Google's Imagen



A chromeplated cat sculpture placed on a Persian rug. Android Mascot made from bamboo. Intricate origami of a fox and a unicorn in a snowy forest.

Midjourney

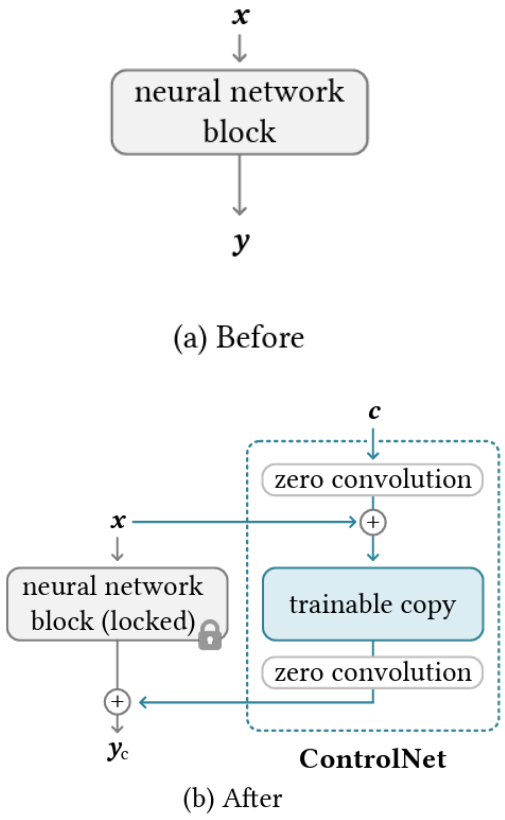


Adobe Firefly



… many more, but many of the models are proprietary …

ControlNet





L. Zhang, A. Rao and M. Agrawala, "Adding Conditional Control to Text-to-Image Diffusion Models," *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, Paris, France, 2023, pp. 3813-3824, doi: 10.1109/ICCV51070.2023.00355.

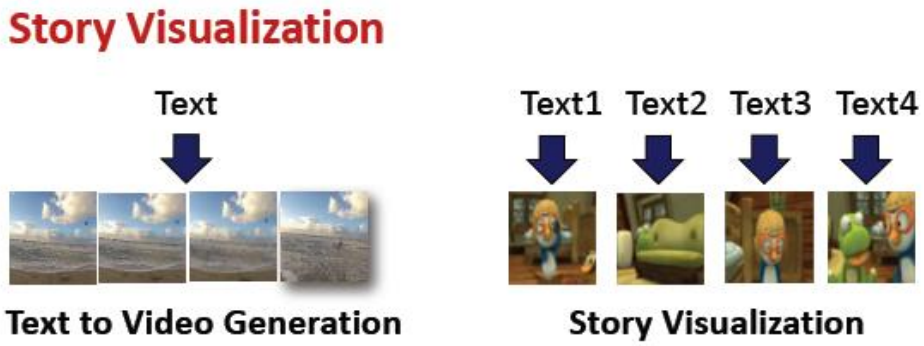
Video GAN

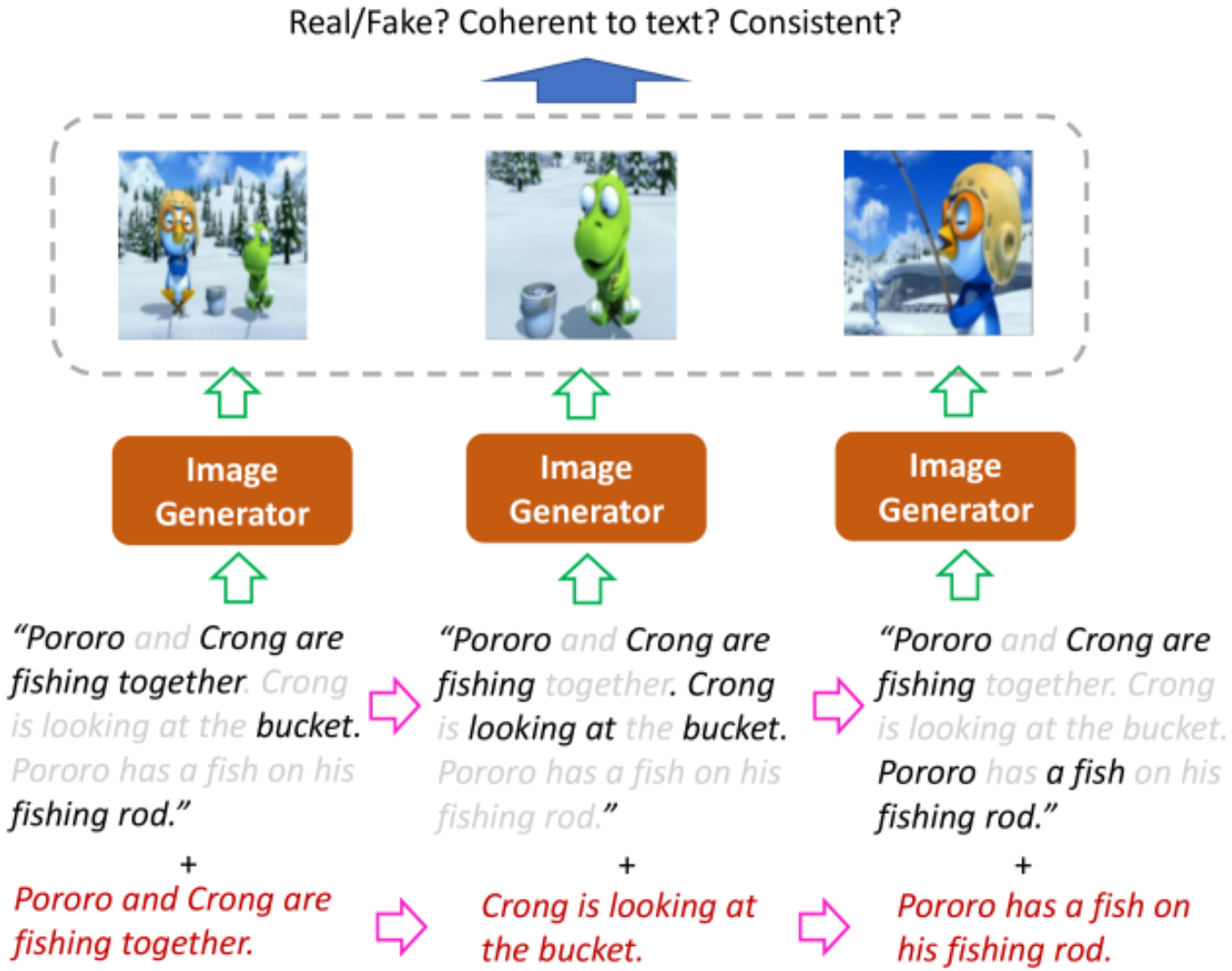
Three main types:

- GANs frameworks that use RNN architectures to address the time-series nature of video data.
- Two-stream architecture video GANs, where each stream considers a different aspect of the video.
- Progressive video GANs models in which initial frames are first generated, and the generated data is fed into another generator to produce an enhanced result.

Text to Video

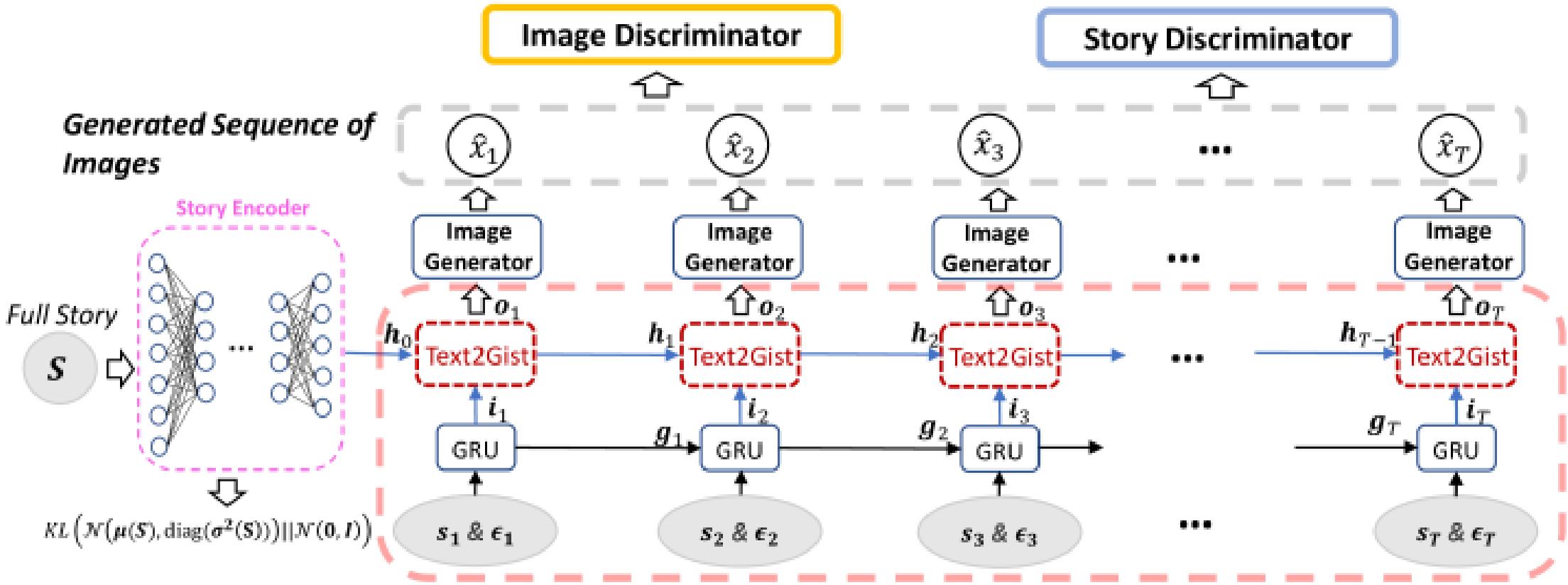
Challenge: The generated image sequence must consistently and coherently depict the whole story and maintain the logic of the storyline





StoryGAN: A Sequential Conditional GAN for Story Visualization

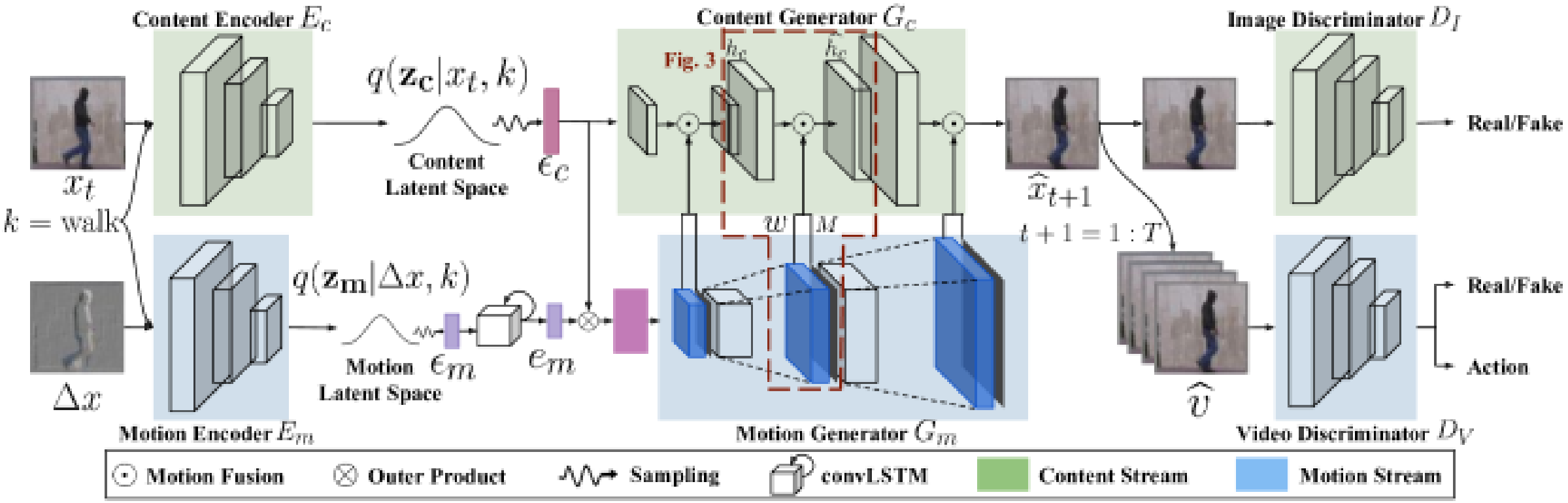
StoryGAN uses a gated recurrent unit (GRU) RNN to incorporate the previously generated images into the current sentence's image generation.



] Y. Li et al., "StoryGAN: A Sequential Conditional GAN for Story Visualization," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019, pp. 6329-6338.

Two-Stream VAN for Video Generation

Generates a video from an action label and a disentangled noise vector



Content Stream + Motion Stream

X. Sun, H. Xu, and K. Saenko, "TwoStreamVAN: Improving Motion Modeling in Video Generation", WACV, 2020.

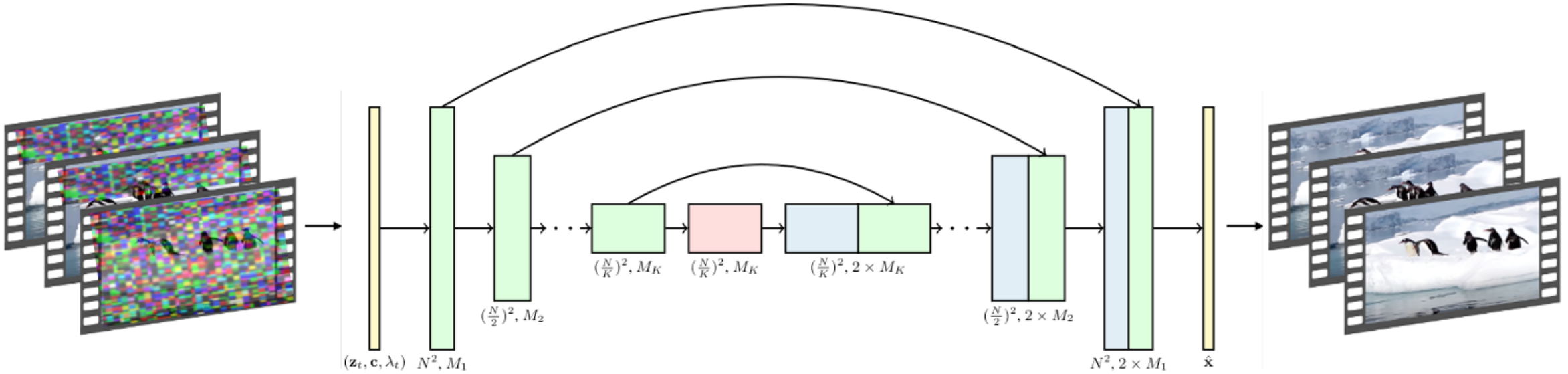
Sora: Video generation models as world simulators

https://openai.com/index/video-generation-models-as-world-simulators/

Transformers, Diffusion model, DeepFake, ….

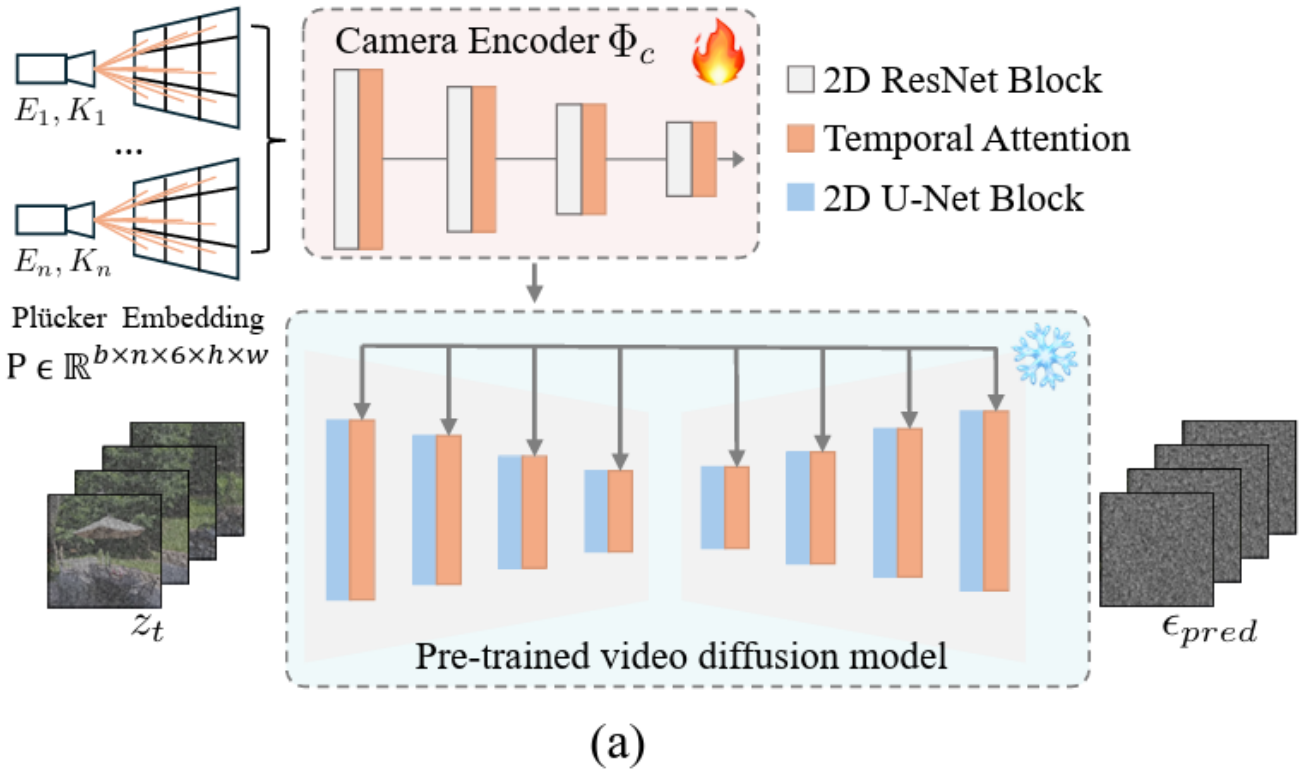
Video Diffusion Model

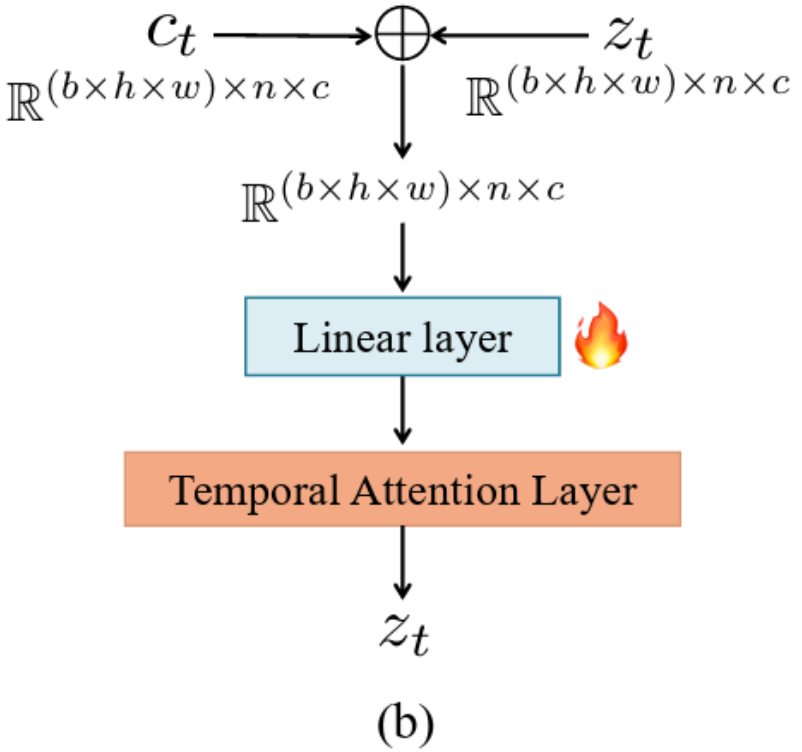
- Architecture = 3D UNet + spatial-temporal attention
- Relative positional embeddings across time axis



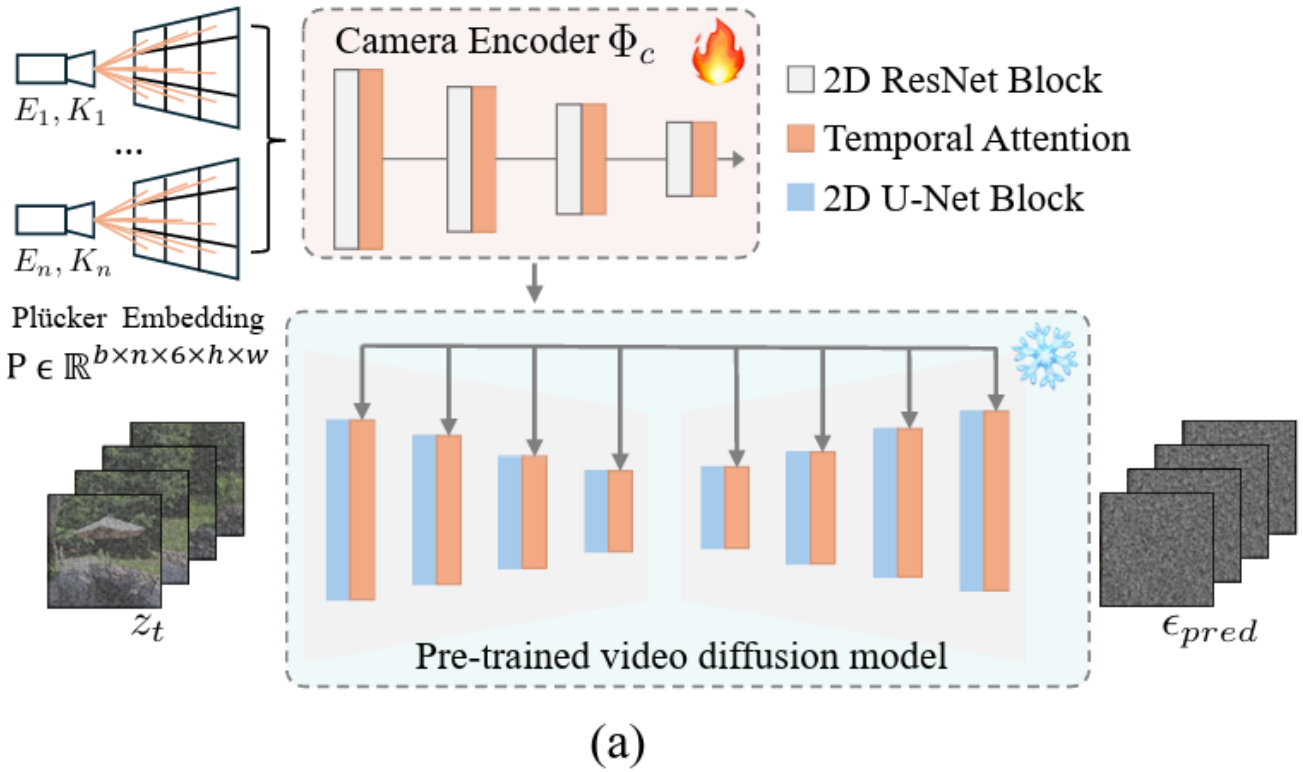
Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. 2022. Video diffusion models. In Proceedings of the 36th International Conference on Neural Information Processing Systems (NIPS '22). Curran Associates Inc., Red Hook, NY, USA, Article 628, 8633–8646.

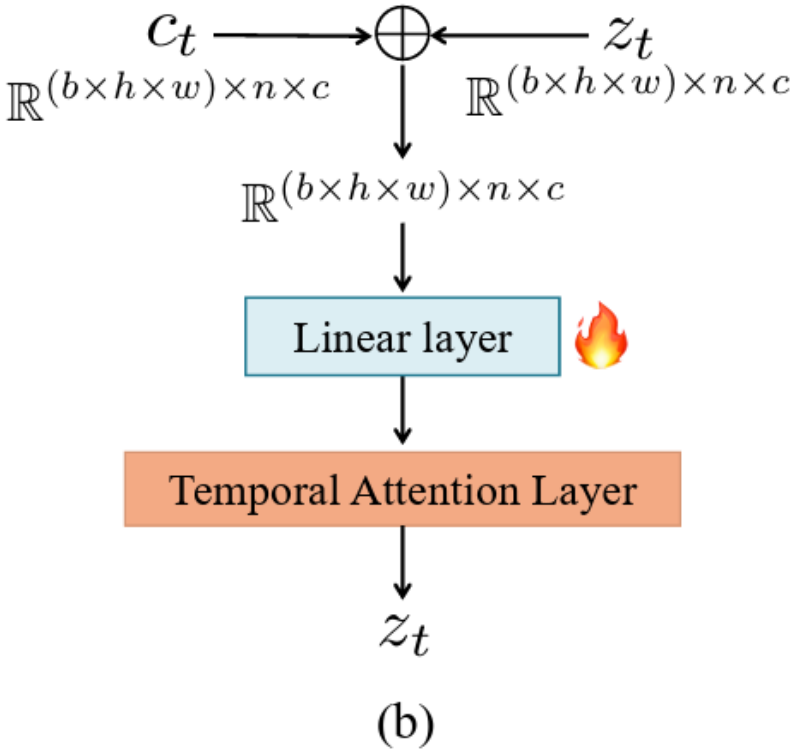
CameraCtrl

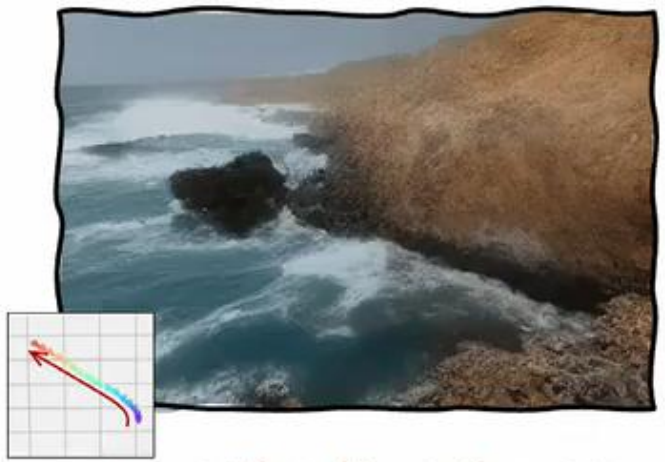

Camera Encoder Φ_c
2D ResNet Block
Temporal Attention
2D U-Net Block
E_1, K_1
...
E_n, K_n
Plücker Embedding
$P \in \mathbb{R}^{b \times n \times 6 \times h \times w}$
z_t
Pre-trained video diffusion model
ϵ_{pred}
(a)


c_t
$\mathbb{R}^{(b \times h \times w) \times n \times c}$
z_t
$\mathbb{R}^{(b \times h \times w) \times n \times c}$
$\mathbb{R}^{(b \times h \times w) \times n \times c}$
Linear layer
Temporal Attention Layer
z_t
(b)

CameraCtrl


Camera Encoder Φ_c
2D ResNet Block
Temporal Attention
2D U-Net Block
E_1, K_1
...
E_n, K_n
Plücker Embedding
$P \in \mathbb{R}^{b \times n \times 6 \times h \times w}$
z_t
Pre-trained video diffusion model
ϵ_{pred}
(a)


c_t
$\mathbb{R}^{(b \times h \times w) \times n \times c}$
z_t
$\mathbb{R}^{(b \times h \times w) \times n \times c}$
$\mathbb{R}^{(b \times h \times w) \times n \times c}$
Linear layer
Temporal Attention Layer
z_t
(b)



Rocky coastline with crashing waves



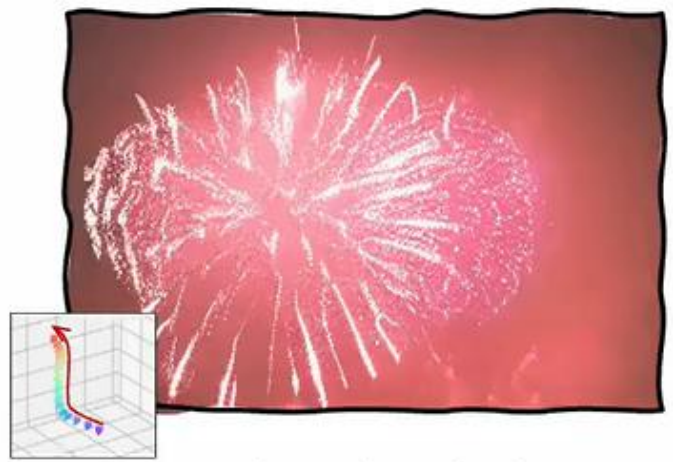
A serene mountain lake at sunrise, with mist hovering over the water.



flowing waterfalls, its cascades forming staircases



A sparse pine forest covered in fresh snow.



A firework display illuminating the night sky.



Natural hot spring the steam floats up.

https://hehao13.github.io/projects-CameraCtrl/

Summary

- GANs are generative models that are implemented using two types of stochastic neural network modules: **Generators** and **Discriminators**.
- **Generator** tries to generate samples from random noise or condition as input
- **Discriminator** tries to distinguish the samples from Generator and samples from the real data distribution.
- Both networks are trained adversarially to fool the other component. In this process, both models become better at their respective tasks.

Assignment

- **Assignment: Designing a Video Generation Network—Your Solution versus an AI Solution**
- Suppose you are tasked with developing a deep learning system that generates synthetic videos depicting Donald Trump for clearly labelled educational or research demonstrations.
- Develop your own proposal, use a generative AI tool to obtain an alternative proposal, and critically compare the two. The aim is to assess your technical understanding and your ability to evaluate AI-generated suggestions.

Assignment

Part 1 Before consulting a generative AI tool, describe your proposed approach, covering:

- **Data:** What data would you need? How would you collect, preprocess, and organise the training and evaluation datasets?
- **Network architecture:** What are the main components, inputs, and outputs? How would your model maintain identity and temporal consistency across frames?
- **Resources:** What hardware, software, pretrained models, and computational budget would you require?
- **Design rationale:** Explain your choices, assumptions, expected limitations, and trade-offs.

Assignment

Part 2: AI-assisted proposal, critical comparison, and revised design

- Use a generative AI tool to propose a solution to the same task and compare the AI-generated proposal with your independent proposal.
- Explain the main similarities and differences in data, architecture, resources, and evaluation.
- Identify at least one weaknesses, unsupported claims, or missing details in the AI proposal. Explain how you checked them.
- Identify at least one useful AI suggestion and explain whether you would adopt, modify, or reject it.

Assignment

Part 3: Conclusion

- How do you think AI can help you develop and improve your proposed solution?

Please submit your work in PDF format, encouraged to be within 5 pages.

Assessment criteria

- Independent proposal and technical reasoning: **30%**
- Critical comparison and verification of AI suggestions: **50%**
- Clarity, transparent reporting of AI use, and responsible practice: **20%**

Online submission system close after 10 Nov 2026